Post Mining Land Use | Rezoning | Macquarie Coal Complex Pilot Project

Utilities Engineering Report

Department of Planning, Housing and Infrastructure

Reference: 531168-000

Revision: B

2026-07-01

aurecon

Bringing ideas to life

Document control record

Document prepared by:

Aurecon Australasia Pty Ltd

ABN 54 005 139 873

Level 11, 73 Miller Street
North Sydney 2060 Australia

PO Box 1319
North Sydney NSW 2059
Australia

T +61 2 9465 5599
F +61 2 9465 5598
E sydney@aurecongroup.com
W aurecongroup.com

Document control						aurecon
Report title		Utilities Engineering Report				
Document code		N/A	**Project number**		531168-000	
Client		Department of Planning, Housing and Infrastructure				
Client contact		Clare Butterfield	**Client reference**		PMLU MCC Pilot Project State-led Rezoning	
Rev	**Date**	**Revision details/status**	**Author**	**Reviewer**	**Verifier** (if required)	**Approver**
A	2026-06-16	Draft	ABO, AW, DM, FB, JB	LR	GL	GL
B	2026-06-30	Final For Exhibition	ABO, AW, DM, FB, JB	LR	GL	GL
Current revision		**B**				

Approval			
Author signature		Approver signature	
Name	Josephine Butler	Name	Greg Lee
Title	Civil Engineer, Water	Title	Practice Lead, NSW Land and Water

Executive summary

This Utilities Engineering Report demonstrates that the Macquarie Coal Complex Precinct can be effectively serviced to support its transition to a post-mining land use and provides a robust technical basis for rezoning. The Precinct benefits from strong regional infrastructure connectivity, particularly in electricity, and access to substantial on-site water resources, creating a high level of inherent servicing potential. While legacy mining infrastructure and fragmented utility networks present constraints, these are considered manageable within a staged delivery framework. Importantly, the site is uniquely positioned to adopt both traditional "business as usual" servicing solutions and alternative, decentralised approaches that align with long-term sustainability and industrial transition objectives.

Water Servicing

The Precinct is well positioned within Hunter Water's regional supply system, with Grahamstown Water Treatment Plant providing **246 ML/d of capacity and forecast spare capacity of ~17.5 ML/d in 2029**, exceeding the Precinct's **ultimate extreme day demand of ~16 ML/d**. This demonstrates that, at a regional scale, potable water supply is not a limiting factor to development.

However, constraints exist at a local level, with limited existing infrastructure across the site, fragmented connections and the need to establish new trunk mains. These constraints are most pronounced in the southern and western areas. The presence of substantial on-site stored water and proximity to recycled water sources presents a key opportunity to supplement potable demand, enabling **precinct-scale alternative water systems (e.g. recycled water, raw water use)**. This creates a pathway to reduce reliance on trunk augmentations and support a more resilient, lower-cost servicing strategy over time.

Wastewater Servicing

Wastewater represents one of the most significant constraints to development. The majority of the Precinct is **not connected to the Hunter Water wastewater network** and historically relied on septic systems. This will require the delivery of entirely new infrastructure including pump stations and rising mains to connect to Edgeworth Wastewater Treatment Works.

Critically, Edgeworth WWTW is projected to **reach capacity by 2029**, with no confirmed timing or funding for upgrades. This introduces a material risk to staging and delivery, as development is partially dependent on external infrastructure upgrades.

At a precinct scale, this constraint strongly aligns with **opportunities for decentralised servicing**, including on-site wastewater treatment, water recycling and reuse systems. These approaches could reduce reliance on regional infrastructure in early stages and unlock development ahead of major augmentation works.

Electricity Servicing

Electricity is a major enabling service for the Precinct. The site has access to multiple voltage levels (11 kV to 330 kV) and is located near key infrastructure including Newcastle BSP and Argenton Zone Substation. Available capacity includes **~30 MVA at the 11 kV level and up to ~400 MW at the 132 kV network level**, providing significant long-term supply potential.

Total precinct demand is estimated at **~104 MVA (low case) to ~190 MVA (high case)**, which can be staged progressively through feeder upgrades, substation augmentation, and potential new zone substations.

Key constraints relate to the need for staged augmentation and potential long-term land take for substations and transmission corridors. However, electricity also underpins a major opportunity to position the Precinct as a **low-carbon industrial hub**, supporting electrification of industry, renewable energy integration, microgrids and energy storage.

Gas and Energy Strategy

There is currently **no gas infrastructure within the Precinct**, with the nearest connection point located at Racecourse Road and Blair Street. This limits feasibility for widespread gas reticulation, particularly in western areas where lead-in distances are significant.

Rather than a constraint, this creates a strategic opportunity to **minimise reliance on gas and prioritise electrification and alternative energy systems**, including renewable generation, battery storage and emerging fuels such as hydrogen. Gas is therefore best treated as a targeted, case-by-case utility for specific users rather than a core servicing backbone.

Telecommunications

Telecommunications is not a fundamental constraint to development but requires staged delivery. Baseline mobile coverage is strong, and connections can be established from existing infrastructure at Powerhouse Road and The Broadway.

Demand remains difficult to quantify at this stage, particularly given the potential for high-data users such as data centres. The servicing approach relies on a **developer-led NBN model**, with pit and pipe infrastructure delivered upfront and fibre rolled out incrementally.

Opportunities exist to develop a **high-quality digital precinct**, with scalable fibre networks, redundancy, and integration of smart infrastructure. Additional mobile towers and bespoke connections for large data users may be required as demand grows.

Key Cross-Cutting Constraint: Cockle Creek

Cockle Creek represents a major physical and servicing constraint across the Precinct. The creek creates a barrier between the site and existing infrastructure networks, requiring new crossings for potable water, wastewater and other utilities.

While feasible, these crossings introduce **significant cost, delivery complexity and approval risk**, particularly where trenchless methods are required. This constraint reinforces the importance of staged delivery and supports the case for **decentralised servicing strategies**, particularly in early development phases.

Overall Conclusion

The assessment confirms that the Precinct can be serviced through a combination of traditional infrastructure augmentation and staged delivery aligned with development uptake. While constraints such as wastewater capacity, legacy infrastructure and Cockle Creek present challenges, they are not prohibitive to development and instead provide a clear framework for infrastructure planning.

Importantly, many of these constraints align with and reinforce the opportunity to adopt **alternative servicing approaches**, including decentralised water systems, electrification, renewable energy integration and scalable telecommunications infrastructure.

On this basis, the report demonstrates that a viable servicing pathway exists to support the proposed land uses and confirms that the Precinct is **suitable for rezoning**, with sufficient flexibility to accommodate both conventional and innovative infrastructure solutions as development evolves.

Contents

Figures

Tables

1

Main report

1 Introduction

This Utilities Engineering Report (this Report) has been prepared by Aurecon on behalf of the Department Planning, Housing and Infrastructure (DPHI) to support the State-led rezoning of the Macquarie Coal Complex for Post Mining Land Use. The Macquarie Coal Complex Land Transformation Precinct (the Precinct) forms part of the NSW Government's investigation into future land use pathways for mining affected land.

The Report outlines a forward-looking servicing strategy for the Macquarie Coal Complex, positioning the Precinct to leverage strong regional utility connectivity and inherent site advantages. The site benefits from proximity to established electricity infrastructure with access to multiple voltage levels and significant available capacity, enabling both staged development and accommodation of high-demand industrial users over time. In addition, the presence of substantial on-site stored water and access to nearby treatment facilities presents a unique opportunity to integrate alternative water sources, supporting more sustainable and resilient servicing outcomes as the Precinct evolves.

These advantages are complemented by the site's long history of mining and industrial use, which has shaped the existing utility environment. Legacy infrastructure, including fragmented potable water connections and limited wastewater servicing, reflects its former operational requirements, with some areas reliant on isolated or off-grid systems. While this presents constraints in its current form, it also provides a clear framework for targeted upgrades and strategic planning, enabling the Precinct to transition from a legacy mining landscape into a well-serviced, diversified employment and mixed-use hub.

The following areas are herein referred to as the site and form the subject area under assessment in this report.

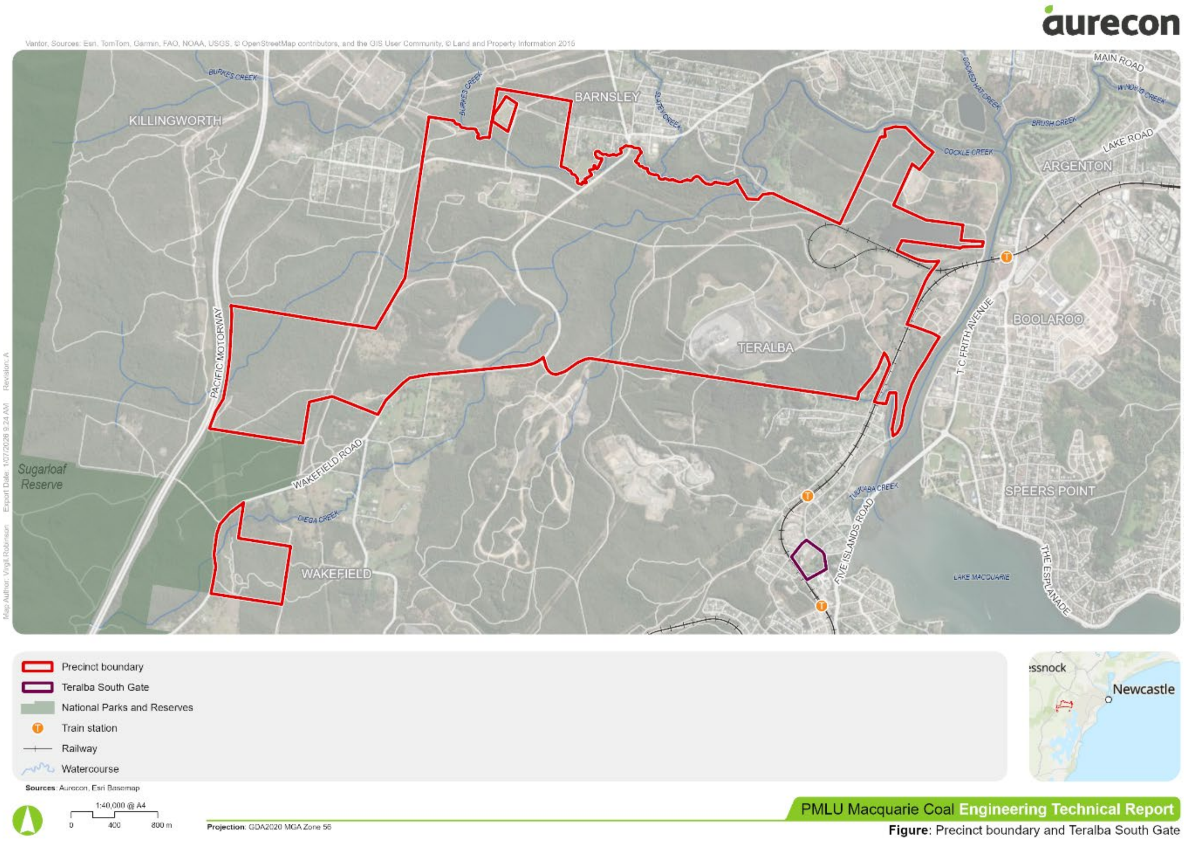

Figure 1-1 Precinct Site Boundary

1.1 Purpose of this report

The purpose of this report is to outline the utility servicing strategy for the Structure Plan of the Precinct. It presents the existing utility infrastructure in the investigation area, highlights constraints and servicing opportunities, provides indicative estimates of demand for utilities based on assumptions regarding land use, and scopes required infrastructure to support its delivery.

The assessment of expected utility demand and associated future utility projects is based on the Structure Plan. The expected utility demands will be conservative assumptions of the requirements of the development due to the nature of the proposed land types in the Structure Plan. The level of accuracy of the expected utility demands will increase as the land use types are refined in future phases of the site planning process and as development opportunities are realised. The Structure Plan preliminary servicing strategy is based on available desktop information and does not replace Authority approval.

The assessment is based on a best estimated land use and uptake over time. It is acknowledged that the this may vary as development is realised and or technology changes. Where a material change occurs, a refresh of this report may be required with an associated master plan amendment.

1.2 Methodology

This report is prepared in a traditional 'business as usual' servicing strategy manner, followed by exploration of alternative servicing options. As a minimum test of adequacy for the purpose of rezoning, the default position is to service the site through traditional means with major utility authorities. Opportunities to utilise existing infrastructure are initially prioritised, this presents the greater and most cost effective leverage to servicing a site. It also assists in initial enabling works to support early development. The resulting strategy aims to ultimately ensure that a viable servicing option is available and therefore supporting the rezoning of development. The following methodology is adopted in this assessment:

- Undertake a desktop review of existing trunk utility provision across the site and within the regional network.
- Assess residual capacity for growth locally and within the regional network
- Evaluate existing infrastructure that may form a physical or site encumbrance constraint to development (such as easements, major infrastructure, high risk infrastructure)
- Based on a preferred masterplan estimate possible demand scenarios for each utility infrastructure typology.
- Review future demand against residual existing infrastructure and determine 'business as usual' augmentations and upgrades to support the development
- Identify alternative servicing strategies for adoption
- Develop a staging plan for infrastructure and any additional performance criteria/development controls.

1.3 Level of detail and exclusions

This report has been prepared by Aurecon Australasia Pty Ltd for the benefit of our client (DPHI) and was prepared for the specific purpose of the utility engineering assessment for the Macquarie Coal Complex. Aurecon has used its reasonable endeavours to ensure that this document is based on information that was current as of the date of the document.

Aurecon's findings represent its reasonable judgments within the time and budget context of its commission and utilising the information available to it at the time.

- The design and engineering assumptions are high level, consistent with a pre-feasibility level of confidence.
- Details of residual capacity in the surrounding networks is limited to available information from the utility service providers.

- Given the fluid nature of planning and utility services, the advice provided is only applicable at the time of preparing the report and must be revised over time.

The following general assumptions are noted, specific to this report:

- This assessment undertaken is high level and strategic in nature to identify key constraints, opportunities, and risks that have a material impact on the development potential of the Precinct. High level assessments were undertaken however no detailed modelling was undertaken.
- Engineering design services beyond the high-level capacity assessment are excluded from this assessment.
- No intrusive or invasive site investigations have been undertaken for this report, any investigations are limited to a review of available desktop information only.
- The advice provided herein remains subject to authority confirmation.

1.4 Reliance on data and information

This Report has been prepared for the benefit of the Client and no other party.

1. Post Mining Land Use – Deliverability of Pilot Projects – Due Diligence Report – Macquarie Coal Complex, prepared by Aurecon in February 2026.
2. Preliminary statutory planning due diligence prepared by Patch Planning.
3. Insights obtained from internal workshops with DPHI, package leads and agencies.

Water Discipline Reliance Documentation

1. Edgeworth Wastewater System Report, Hunter Water (2025)
2. Toronto Wastewater System Report, Hunter Water (2025)
3. South Wallsend and West Lake Macquarie Water System Report, Hunter Water (2024)
4. Independent Environmental Audit, West Wallsend Colliery (2022)
5. Hunter Water Asset Locations (GIS)
6. Wastewater Network, Hunter Water (infographic)
7. Water Network, Hunter Water (infographic)
8. Gravity Sewerage Code of Australia WSA 02—2014-3.1 Hunter Water Edition Version 2
9. Water Supply Code of Australia WSA 03—2011-3.1 Hunter Water Edition Version 2

Electricity Discipline Reliance Documentation

1. Report Data for ZN00224 Argenton 132/11, Ausgrid - Rosetta Data Portal (2026)
2. Report Data for Newcastle 132 kV BSP, Transgrid - Rosetta Data Portal (2026)
3. AS/NZS 3000:2018 Electrical installations (2018)
4. Look Up and Live Tool, Before You Dig Australia (2026)
5. Easement Guidelines, Transgrid (2024)
6. Development Industry Briefing, Citipower & Powercor (2024)

Waste Discipline Reliance Documentation

Documents informing waste demand estimates are referenced where they were applied (see section 3.7.1).

2 Baseline analysis

The baseline analysis outlines the existing utility networks within and surrounding the Precinct to support an understanding of how the proposed development can be effectively serviced. A desktop review of available data was undertaken to identify current network infrastructure and capacity. This assessment also informed the key servicing opportunities and constraints relevant to future planning.

The investigated existing service providers in the area are outlined in Table 2-1.

Table 2-1 Utility Providers Operating in the Precinct and Surrounding Area

Service	Service Provider
Potable Water and Wastewater	Hunter Water Corporation
Electricity	Transgrid
Electricity	Ausgrid
Telecommunications	Telstra
Telecommunications	Optus and/or Uecomm
Telecommunications	Nextgen cable
Telecommunications	TPG
Gas	Jemena Gas North
Gas	Ampol

2.1 Water and wastewater systems

This section provides a strategic, desktop-based assessment of existing potable water and wastewater servicing conditions, constraints and opportunities within the study area. It also provides recommended actions for post-exhibition phases. The assessment considers the Hunter Water network, existing potable water and wastewater infrastructure, treatment capacity, potential connection points, transfer network capacity, recycled water opportunities and servicing requirements for the Main, Wakefield and Teralba Southgate study areas. The section provides details of the following:

- The existing potable water and wastewater servicing conditions across the study area.
- Overview of opportunities to connect into existing potable water and wastewater infrastructure
- Key constraints and risks associated with potable water and wastewater servicing.
- Recommended actions before and after exhibition to support future precinct planning and development.

2.1.1 Potable water

Current Conditions

The Macquarie Coal Complex Precinct sits within Hunter Water's water supply network and services all current potable water pipelines within the Precinct.

The Precinct sits within Hunter Water's South Wallsend and West Lake Macquarie System (Figure 2-1) which receives flows from Grahamstown Water Treatment Plant. Grahamstown Water Treatment Plant can currently treat 246 ML/d of normal quality raw water; it is expected to have spare capacity beyond 2034. The expected future spare capacity at the Grahamstown WTP is highlighted in Table 2-2.

Table 2-2 Hunter Water's Grahamstown Water Treatment Plant Expected Growth and Spare Capacity based on 246 ML/d Treatment (normal raw water quality) (Hunter Water, 2024)

System Summary	Projected		
	2024	2029	2034
Total Equivalent Population (EP)	430,000	455,000	478,000
Average Day Demand (ADD) (ML/d)	124.5	121.9	122.9
Max Day Demand (MDD) (ML/d)	221.1	228.5	233.3
Spare Capacity at MDD (ML/d)	24.9	17.5	12.7

Within the South Wallsend and West Lake Macquarie System, the Precinct falls within Elermore Vale 1 Reservoir System (Figure 2-1) which is supplied by a DN1050 main from a water pump station in Wallsend. The Elermore Vale site has two existing reservoirs which gravity feed the system, based on desktop investigations these reservoirs were found to have a combined capacity of 110 ML (60 ML and 50 ML).

The existing Hunter Water potable water network in and around the Precinct is highlighted in Figure 2-2.

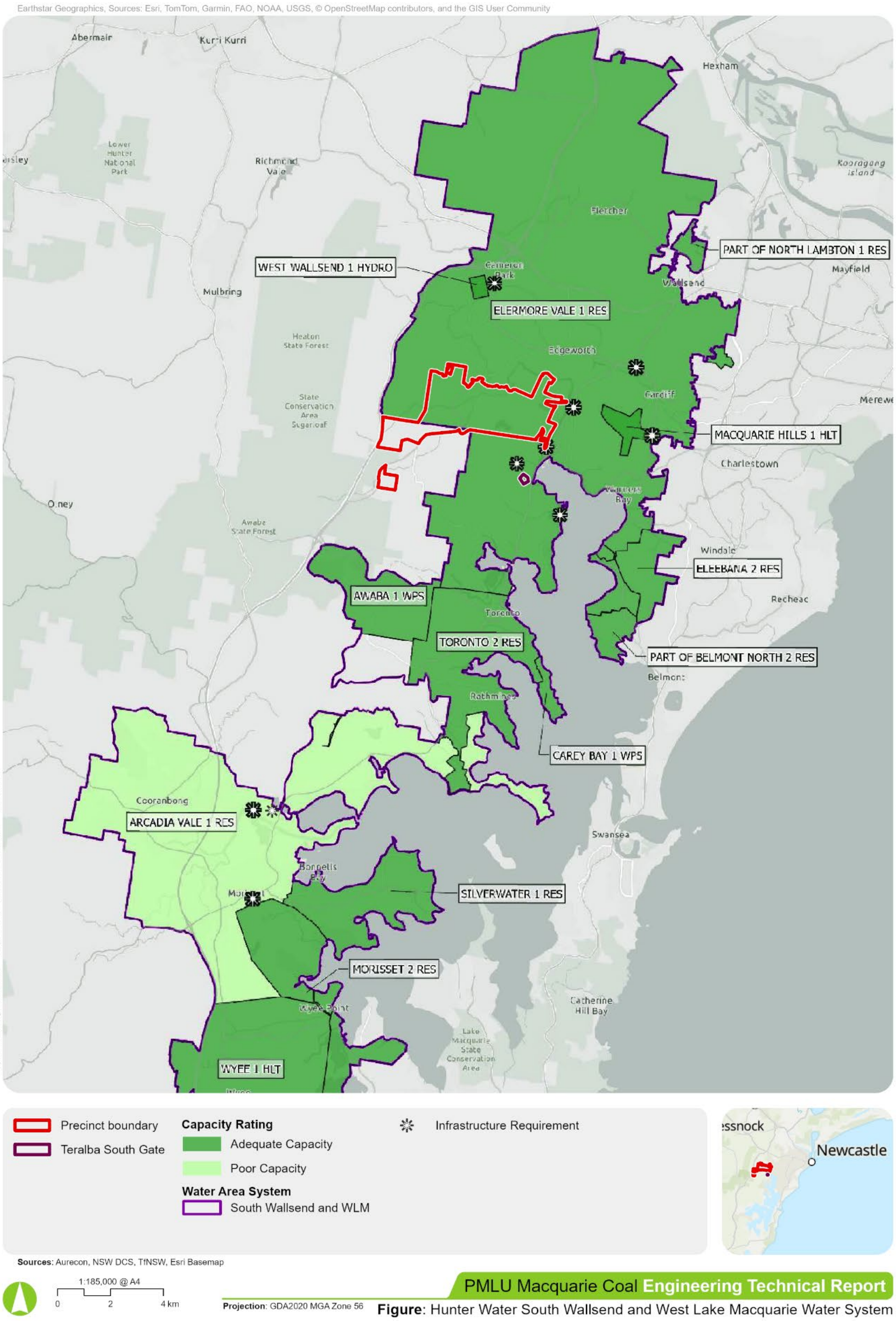

Figure 2-1 Hunter Water South Wallsend and West Lake Macquarie System (Hunter Water, 2024)

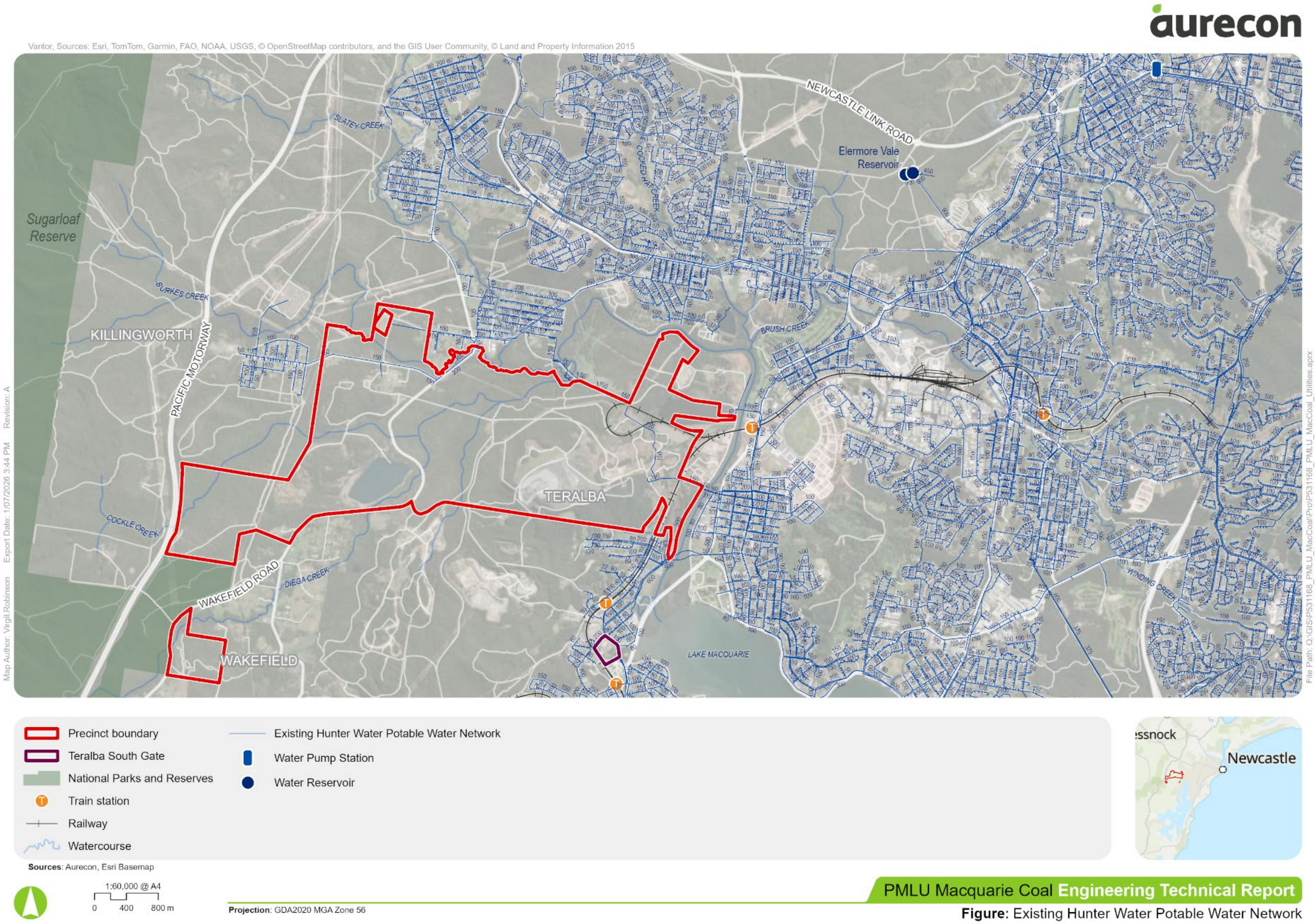

Figure 2-2 Existing Hunter Water Potable Water Network

Due to the large extents of the Precinct, the current conditions, opportunities, constraints, and risks of the potable water network will be broken down into three separate areas:

- Main Precinct Study Area,
- Teralba Southgate Study Area, and
- Wakefield Study Area.

See Figure 2-3.

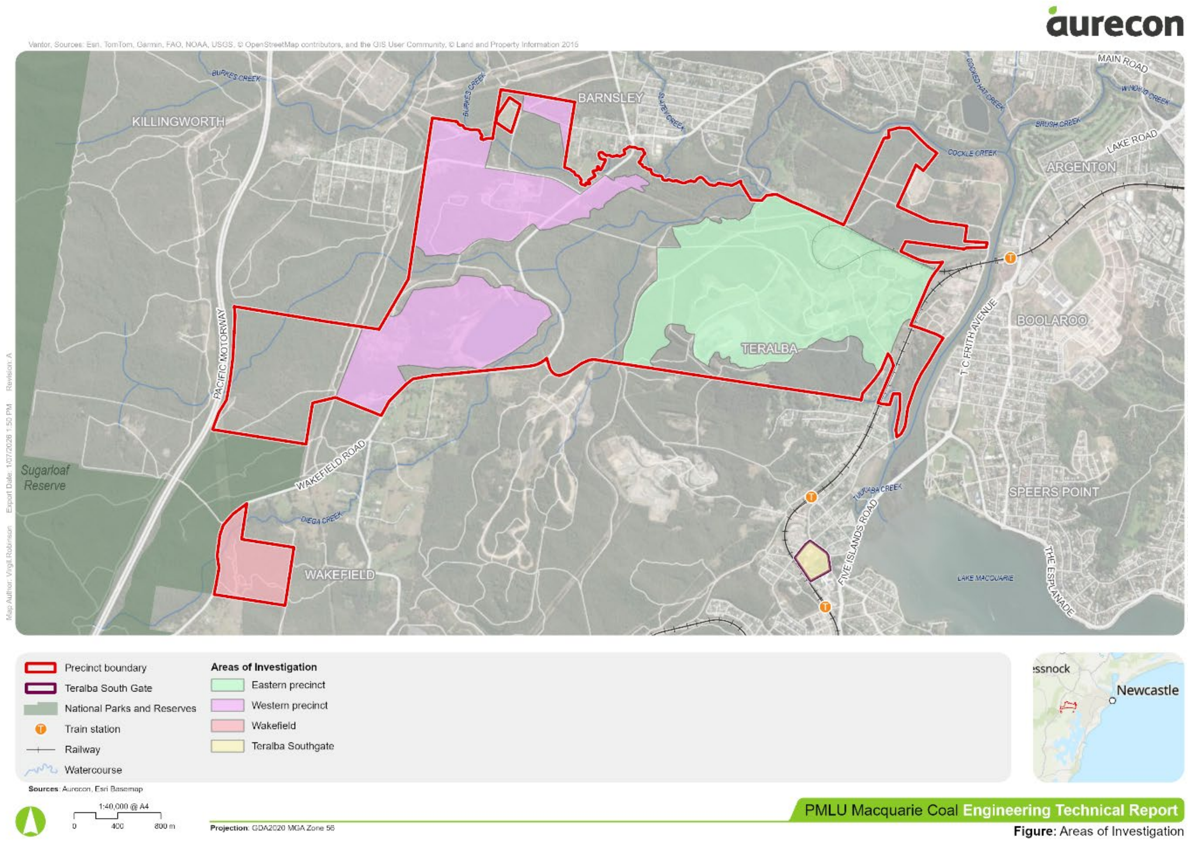

Figure 2-3 Potable Water and Wastewater Areas of Investigation

Main Precinct Study Area

The northern section of the Precinct is supplied via two separate DN150 potable water pipelines. To the east, a DN150 crosses Cockle Creek from Boolaroo and runs through the study area, crossing the railway loop and running along The Weir Road into Barnsley. At the western side of the site, a DN150 runs along The Broadway and Wakefield Road between Barnsley and Killingworth, this pipeline currently services the existing site office off The Broadway. The southern section of the Precinct is largely isolated from the potable water network. The existing potable water network is shown in Figure 2-2.

Teralba Southgate Study Area

The Teralba Southgate study area is currently serviced by Hunter Water's potable water network via an unknown pipe size which branches off a DN150 along York Street. The network within the site is unknown. The surrounding streets have multiple potable water pipelines ranging from DN150-600. The existing potable water network is shown in Figure 2-2.

Wakefield Study Area

The Wakefield site has no connection to the Hunter Water network, nor does it have any Hunter Water assets near the site. There are no other known sources of potable water in this area, and it is known that the nearby residential properties are off grid for their water supply.

Opportunities

The projected spare capacity at Grahamstown Water Treatment Plant through to 2034 presents an opportunity for the Precinct to leverage some of this spare capacity to service the proposed development.

The Precinct benefits from having existing infrastructure to the north, Killingworth, Barnsley and Edgeworth, and to the east, Boolaroo and Argenton. This provides an opportunity to tap into these networks to reduce the need for significant infrastructure investment, this is dependent on the spare capacity within these networks which is unknown at this stage. The existing pipeline crossings from Boolaroo, DN315 and DN600, from which the DN150 through the Precinct branches off from, presents a potential opportunity to connect into one of the larger pipes to utilise the existing crossings of Cockle Creek.

The Teralba Southgate study area is well positioned for future potable water growth needs, the site has an existing connecting into the Hunter Water network and is located in close proximity to larger pipelines if larger volumes of water are required.

There is an opportunity to reduce the reliance on the potable water network by utilising nearby raw or recycled water sources to supplement potable water supply where land uses do not require potable quality water. Recycled water from Edgeworth Wastewater Treatment Works presents a potential supply opportunity, however, this should be further investigated to understand water quality and volume. Waratah Golf Course in Argenton currently uses effluent flow from the plant to irrigate their land. The large volumes of stored water onsite also present an opportunity to reduce potable water requirements, again the quality of water will need to be further investigated to determine if this option is feasible. Both of these opportunities are dependent on the proposed land uses for the Precinct.

Constraints and Risks

A large risk to the potable water servicing of the Precinct is the unknown capacity within the Hunter Water potable water transfer networks. At this stage, assumptions have been made as to feasible connection points, however, these assumptions are susceptible to change once greater engagement with Hunter Water is undertaken to better understand the potable water network in the surrounding area.

There are large areas of the Precinct that are not currently serviced by Hunter Water, nor are located near an existing network. These areas include Wakefield study area and the southern section of the main Precinct. The barriers to connect these areas into the network will be greater and may restrict or delay their development potential.

Cockle Creek, which surrounds much of the Precinct, produces a large geographical constraint as new infrastructure is likely required to cross the creek to bridge between the existing networks and the Precinct. Any potential creek crossings, whist feasible, present a higher cost for infrastructure construction.

The existing pipelines within the Precinct are small and unlikely to have spare capacity; it is assumed that these will not be able to support significant development within the Precinct.

Recommendations

- An application to Hunter Water has been made for the Precinct and any amendments to the strategy will be captured prior to finalisation. The below have been flagged with Hunter Water,
  - Investigate tapping into Cockle Creek crossing at the end of Creek Reserve Road where the DN600 & DN315 cross
  - Understand spare capacity within the transfer network in the area

– Confirm potable water capacity from Grahamstown WTP

Post-exhibition

- Staging of infrastructure
- Co-funding opportunities

2.1.2 Wastewater

Current Conditions

The Precinct is not connected into but is surrounded by Hunter Water's wastewater network. Hunter Water service all infrastructure around the Precinct and at the Teralba Southgate site.

The existing Hunter Water wastewater network around the Precinct is highlighted in Figure 2-4.

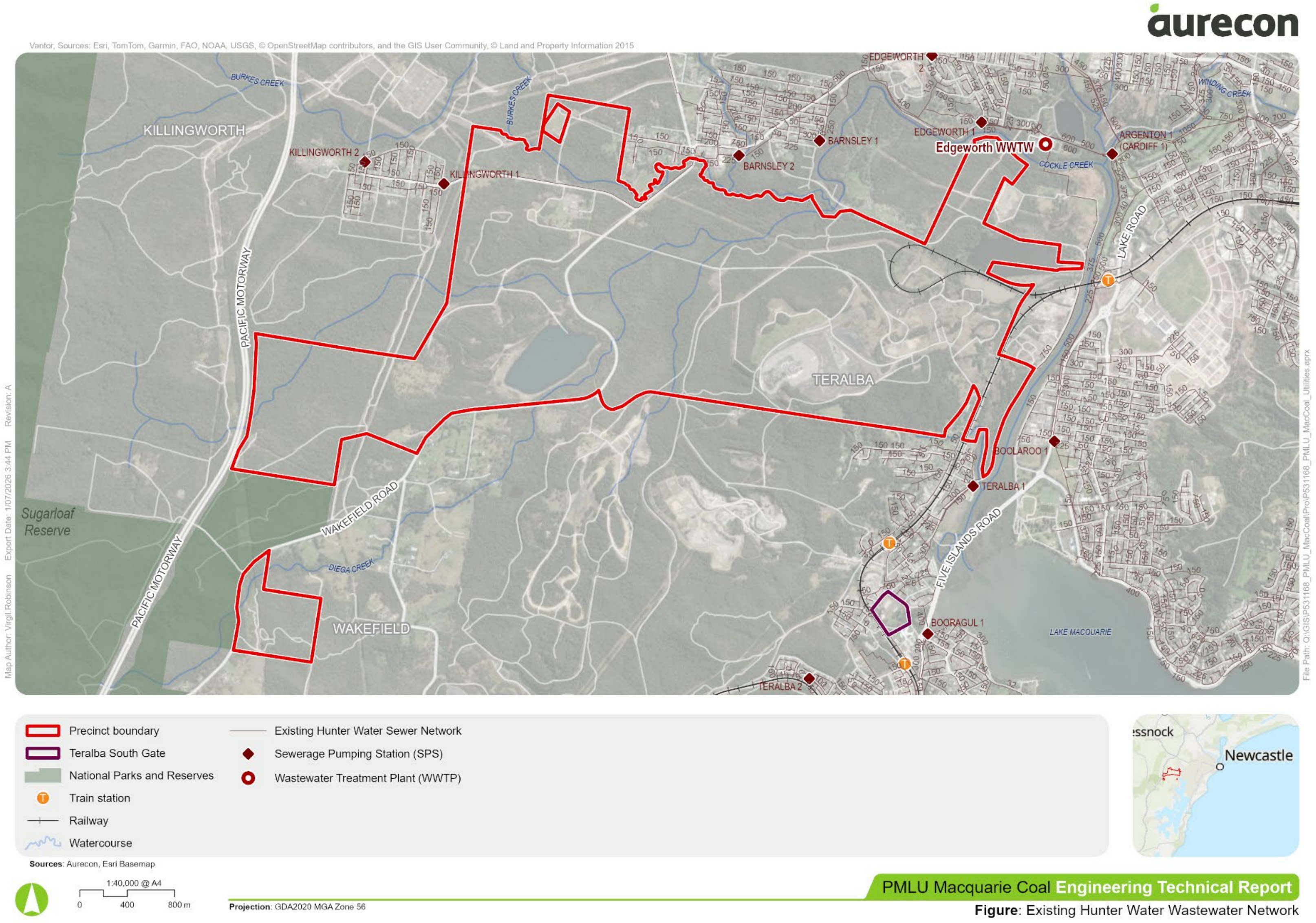

Figure 2-4 Existing Hunter Water Wastewater Network

Due to the large extents of the Precinct, the current conditions, opportunities, constraints, and risks of the potable water network will be broken down into three separate areas:

- Main Precinct Study Area,
- Teralba Southgate Study Area, and
- Wakefield Study Area.

See Figure 2-3.

Main Precinct Study Area

Although it sits outside of any existing sewer system areas, the Precinct is located within close proximity to Edgeworth Wastewater Treatment Works. It is assumed that future wastewater infrastructure within the Precinct would connect into Edgeworth Wastewater Treatment Works.

Edgeworth Wastewater Treatment Works has a current treatment capacity of 70,000 EP or 14 ML/d, it is expected that the plant will reach capacity by 2029. The expected future spare capacity at the plant is highlighted in Table 2-3. Whilst upgrades are planned for the site, timing is unknown and no funding is currently allocated.

Table 2-3 Hunter Water's Edgeworth Wastewater Treatment Works Expected Growth and Spare Capacity based on 70,000 EP treatment capacity (Hunter Water, 2024)

System Summary	Projected		
	2024	2029	2034
Total Equivalent Population (EP)	65,985	70,638	75,293
Average Dry Weather Flow (ML/d)	13.2	14.1	15.1
Spare Capacity at ADWF (ML/d)	0.8	-0.1	-1.1

Note:

- 1 EP was found to be approximately 200 L/d based on projected EP and ADWF

There is currently no wastewater infrastructure within the Precinct, the site was previously serviced by septic tanks during mining operations. No connections into the Hunter Water network are currently present on the site and any future wastewater needs would trigger the need for a wastewater connection from the Precinct into the Hunter Water network, unless flows are to be managed onsite. An existing, decommissioned DN750 MSCL pipeline passes to the east of the site, it transports effluent flows from Edgeworth WWTW, it cannot be utilised to transport sewer flows. The existing wastewater network is shown in Figure 2-4.

Teralba Southgate Study Area

The Teralba Southgate Study Area is currently connected into Hunter Water's network via a DN150 pipeline which crosses York Street into the site. This section of the Precinct sits within the Booragul 1 Sewer System Area which flows through to Toronto Wastewater Treatment Works. Toronto Wastewater Treatment Works has a current treatment capacity of 42,000 EP (8.4ML/d), and it is expected to have capacity beyond 2034. The expected future spare capacity at Toronto WWTW is highlighted in Table 2-4.

Table 2-4 Hunter Water's Toronto Wastewater Treatment Works Expected Growth and Spare Capacity based on 42,000 EP treatment capacity (Hunter Water, 2025)

System Summary	Projected		
	2024	2029	2034
Total Equivalent Population (EP)	38,425	39,908	41,600
Average Dry Weather Flow (ML/d)	7.7	8.0	8.3
Spare Capacity at ADWF (ML/d)	0.7	0.4	0.1

Note:

- 1 EP was found to be approximately 200 L/d based on projected EP and ADWF

Other than the existing pipeline connection into the site, a DN400 gravity pipeline flows from Lake Crescent to Booragul 1 Sewer Pump Station and is the nearest feasible connection point. A DN750 runs to the north and west of the site, it is an effluent pipe from Edgeworth WWTW and cannot be connected into. The existing wastewater network is shown in Figure 2-4.

Wakefield Study Area

Wakefield Study Area has no existing connections to the Hunter Water wastewater network and sits outside of any current sewer system areas. The site is not located in close proximity to any known wastewater networks or sewer system areas. It is known that the nearby residential properties in Wakefield are not connected into the Hunter Water network and have off grid wastewater systems.

Opportunities

The close proximity of the main site to Edgeworth Wastewater Treatment works presents a lower barrier of entry to connect the Precinct into the plant.

Teralba Southgate study area is well connected to the wastewater network with other potential nearby connections if the current pipeline cannot handle projected flows. The treatment plant at Toronto has some spare capacity to take on additional flows which is likely to enable development at the Teralba Southgate site.

Constraints and Risks

A large risk to the wastewater servicing of the Precinct is the unknown capacity within the Hunter Water wastewater networks. At this stage, assumptions have been made as to feasible connection points, however, these assumptions are susceptible to change once greater engagement with Hunter Water is undertaken to better understand the wastewater network in the area.

Edgeworth WWTW is expected to reach capacity before 2029, upgrades are planned for the site, but timing is unknown and there is no current funding approval.

There are large areas of the Precinct that are not currently serviced by Hunter Water, nor are located near an existing network. These areas include Wakefield study area and the southern section of the Precinct. The northern section of the Precinct, whilst having no existing network, is located closer to Edgeworth WWTW. The barriers to connect these southern areas into the network will be greater and may restrict or delay their development potential.

Cockle Creek, which surrounds much of the Precinct, produces a large geographical constraint as new infrastructure is likely required to cross the creek to bridge between Edgeworth WWTW and the Precinct. Any potential creek crossings, whist feasible, present a higher cost for infrastructure construction.

Recommendations

- An application to Hunter Water has been made for the Precinct and any amendments to the strategy will be captured prior to finalisation. The below have been flagged with Hunter Water,
  - Confirm capacity at Toronto Wastewater Treatment Works
  - Confirm capacity at Edgeworth Wastewater Treatment Works and status of upgrades
  - Understand spare capacity within the network in the area
- Investigate wastewater transfer via Barnsley (new road bridge) or new Cockle Creek crossing
- Assess non-potable water reuse and effluent recycling opportunities

Post-exhibition

- Staging of infrastructure
- Co-funding opportunities

2.2 Electricity

This section provides a strategic, desktop-based assessment of existing electrical infrastructure, constraints and opportunities within the Precinct. It also provides recommended actions for post-exhibition phases. The assessment considers nearby transmission and distribution infrastructure, potential connection options, spare network capacity, likely upgrade requirements and the broader regional energy projects. Table 2-5 provides details of the following:

- The existing electricity servicing conditions across the Precinct.
- Overview of opportunities to connect to existing electrical infrastructure.
- Key constraints and risks associated with electricity servicing.
- Recommended actions before and after exhibition to support future precinct planning and development.

A summary of connection levels and their associated risk ratings is provided in Table 2-6 (located toward the end of Section 2.2).

Table 2-5 Electricity Analysis Summary

Electricity	Description
Current conditions	▪ There is sufficient electrical infrastructure to connect to up to the 330kV voltage level at all three “High Development Potential” sites. The remaining areas identified as “further investigation required” have a similar burden. ▪ There also exists several options for connection to nearby 11kV, 33kV and 132kV voltage levels, depending on ultimate load and staging plan. “Future investigation” site area 4 is burdened by two 132kV traversing north south (west of Wakefield Road). ▪ Site has access to multiple voltage levels through Newcastle Substation (330 kV/132 kV), Edgeworth Switching Station (33 kV), and Argenton Substation (11 kV/33 kV/132 kV).
Opportunities	▪ Spare capacity exceeds 20MW at 11/33kV, 1,000 MW at 330 kV and 400 MW at 132 kV. However, it may require the installation of an additional feeders, transformer and/or equivalent upgrade to connect into existing electrical infrastructure. ▪ The site is located nearby to several major transmission and energy projects which would increase existing capacity in the future, these include: – Clean Energy Precinct (CEP): The site is 15km away from the precinct, a leading production, storage, and export hub for hydrogen and ammonia and other clean technologies. – Hunter Central Coast REZ (HCC REZ): The site is located within the HCC REZ, which will bring 1GW of renewable generation to the region.

Electricity	Description
	– Hunter Transmission Project (HTP): 110km of new 500kV transmission lines, which will support up to 8GW of transmission in the Hunter region. ■ The CEP provides 5800 new jobs created in the warehousing / logistics hub which represents opportunities to provide complementary industries or infrastructure in the Precinct Master Plan. ■ Newcastle Substation is a key node in the 330kV network for the Hunter region. With more renewable energy projects coming online, there is potential for the site to provide REZ support by connecting into this 330kV backbone. This can come in the form of supporting infrastructure such as BESS or substations, and ancillary equipment in the Precinct Master Plan.
Constraints and risk	■ To accommodate a load of >100MVA, the proponent will have to look to connect into the 132kV or 330kV bus of Newcastle Substation. This would result in up to 8km of additional transmission, depending on proximity to the electricity asset. ■ 11 and 33kV connection options are limited by bay availability, ability for expansion at the respective substations, and upstream network availability (i.e. feeder and transformer ratings / utilisation). Consultation with relevant NSPs is required to understand feasibility of each connection. ■ Large loads (>100 MVA) likely require connection to 132 kV or 330 kV; limited by spare bay availability and upstream infrastructure. Connection to 330 kV Newcastle Substation requires ~ 8 km of transmission lines with multiple crossings. ■ Consultation with Transgrid/Ausgrid required to confirm feasibility and network upgrades.
Recommendations	
Post-exhibition and as development progresses	■ Consult with Ausgrid and Transgrid for substation availability and connection requirements. ■ Undertake detailed electrical infrastructure assessment to estimate HV network capacity. ■ Align with regional energy projects (REZ, HTP, Port of Newcastle CEP) to identify complementary opportunities. ■ Confirm connection works required and staging implications for each developable area.

Table 2-6 Summary of Connection Levels and Associated Risk Ratings

Connection Level	Distance	Capacity	Risk Rating	Suitable Uses
11 kV	On-site; requires upgrade	< 1 MVA	Low	Light commercial; residential
33 kV	1.5 – 4 km	< 100 MVA	Low–Medium	Moderate industrial
132 kV	1.5 – 3 km	< 500 MVA	Low–Medium	Heavy industry; precincts
330 kV	3 – 8 km	> 500 MVA	Medium	Large-scale generation; transmission-connected

2.3 Waste management

The Precinct falls within the boundaries of the Lake Macquarie City Council, which would be responsible for waste management in the region. This Council currently provides domestic waste services to around 225,000 residents, as well as rate-paying commercial businesses, who are issued with kerbside collection bins for general waste, recycling and organics that are processed at the Awaba Waste Management Facility. Nearby is another Council-owned landfill, being the Summerhill Waste Management Centre (within the City of Newcastle). These are complemented by a range of privately operated waste management facilities operating within a 50 km radius of the Precinct.

A map of existing infrastructure, highlighting the location of the two Council waste management facilities and the region where many of the key private contractors are based, is provided in Figure 2-5.

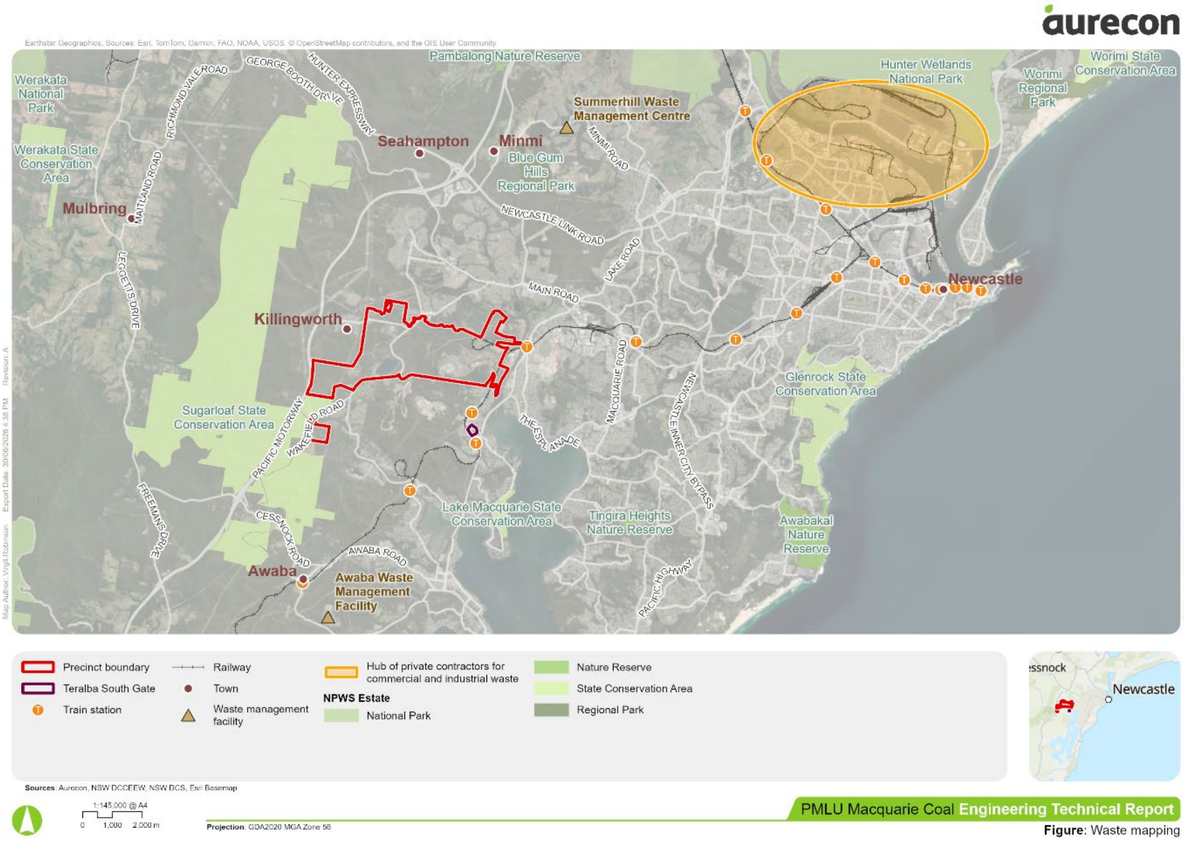

Figure 2-5 Locations of Existing Waste Infrastructure

2.4 Telecommunications, internet and gas

This section provides a strategic, desktop-based assessment of existing telecommunications, internet and gas conditions, constraints and opportunities within the study area. It also provides recommended actions for post-exhibition phases. The assessment considers mobile coverage, nearby telecommunications networks, existing fuel and gas pipeline infrastructure, and the availability of digital and utility services to support future employment and industrial land uses. Table 2-7 provides details of the following:

- The existing telecommunications, internet and gas servicing conditions across the study area.
- Overview of opportunities to support future digital connectivity in the region.
- Key constraints and risks associated with telecommunications, internet and gas servicing.
- Recommended actions before and after exhibition to support future precinct planning and development.

Table 2-7 Telecommunications, Internet and Gas Analysis Summary

Telecommunications, internet and gas	Description
Current conditions	▪ 4G/5G mobile coverage across sites

Telecommunications, internet and gas	Description
	▪ Ampol major fuel pipeline through the northern section of the Precinct; along The Broadway, The Weir Road, and through bushland ▪ Nextgen cable and TPG networks duct crossing the north-west of the Precinct ▪ Jemena high pressure gas pipeline to the west of the Precinct
Opportunities	▪ Teralba Southgate study area is in proximity and connected to services ▪ Good mobile coverage supports digital connectivity ▪ Telecommunications networks in close proximity to the site ▪ Potential to extend network within site to support digital infrastructure requirements
Constraints and risk	▪ Ampol fuel pipeline offset requirements ▪ Lack of telecommunications infrastructure within the study areas ▪ No existing gas network in or around the site
Recommendations	
Post-exhibition and as development progresses	▪ Undertake a Land Use Change Safety Management Study at the DA stage when working near the Ampol pipeline and land uses become clearer

Note: Minor network augmentations are generally required to meet future demand which includes potential augmentation to the nearby fibre access nodes, underground pit and pipe and ducting works. However, these works are generally reactive and targeted as formal customer requests are submitted. They have minimal implications to be considered when developing the Precinct Master Plan as works are generally within the existing and proposed road network.

2.4.1 Gas (Jemena)

Current Conditions

There is no existing gas supply within the study area. The closest Jemema gas main to the site is at the corner of Racecourse Road and Blair Street south east of the site. This 160mm main (operating at 210kPa) could be extended to the site to provide supply for new customers. Gas is a non-essential service and new connections would be on a case by case basis in coordination with Jemena and ultimately based on project feasibility/viability.

The Teralba Southgate site has good access to gas to the north, with connections possible to the 210kPa system.

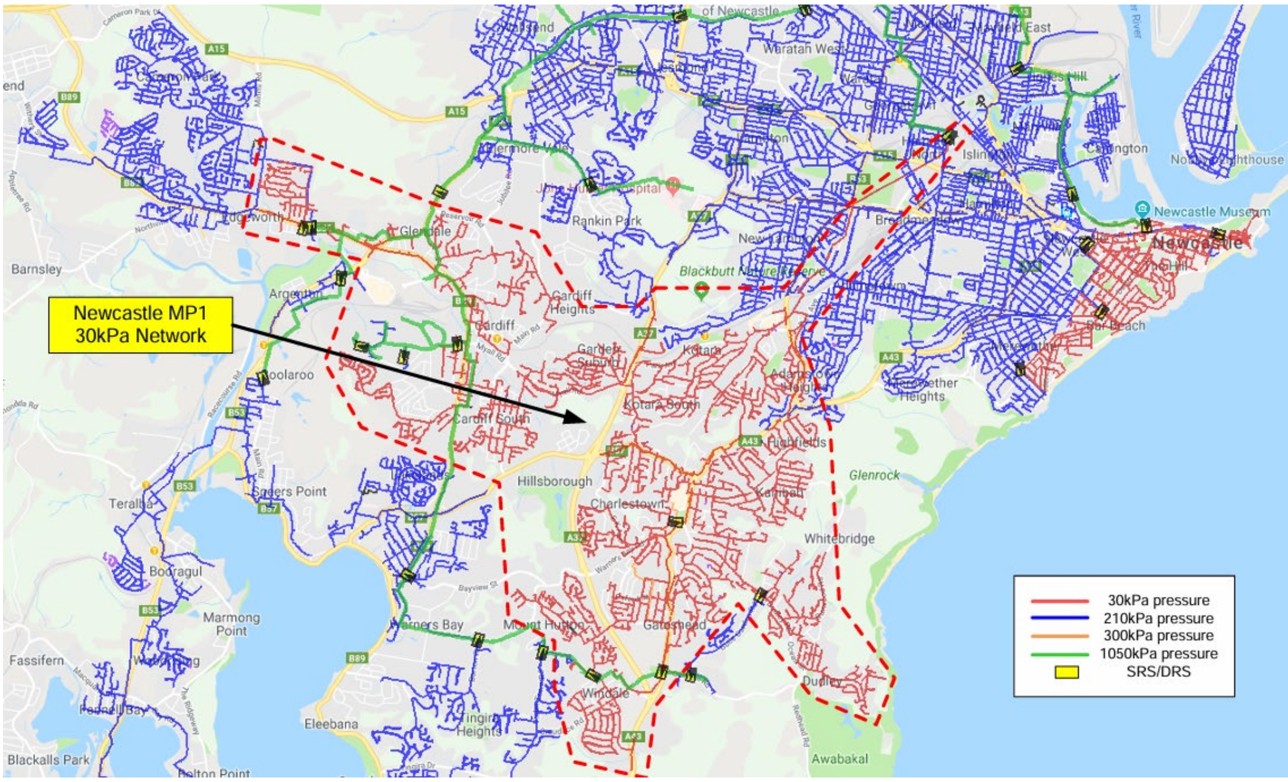

Figure 2-6 Jemena Newcastle Services Map (Jemena, 2023)

Opportunities

Future gas connection could be made to the Blair Street location for the site. It would be encouraged to locate future gas users to the eastern side of the Precinct to capitalise on proximity to this connection point.

The lack of gas does help reinforce exploration of non-gas related and renewable sources of energy within the precinct.

The Teralba Southgate site has good access to gas and could be considered for development of this site.

Constraints

Heavy gas users locating in the western portion of the site may find lead-in works more challenging or commercially unviable without sufficient critical mass.

The majority of precinct users will likely need to connect to all energy through electrification creating a higher burden on electrical usage.

2.4.2 Fuel (Ampol)

Current Conditions

A high-pressure liquid petroleum products pipeline crosses the northern portion of the Precinct. Based on the current utilities mapping, the pipeline generally follows the northern part of the study area along The Broadway, The Weir Road and through adjoining bushland. The pipeline is owned by Newcastle Pipeline Company Pty Ltd and operated by Ampol Petroleum Pty Ltd. It forms part of the Sydney-to-Newcastle fuel supply system, carrying refined petroleum products from Ampol’s Banksmeadow Fuels Terminal to fuel terminals between Sydney and Newcastle. Ampol’s BYDA response identifies that the pipeline transfers approximately 4 billion litres of fuel each year, reducing the need for approximately 45,000 road tanker movements per year between Sydney and Newcastle.

The pipeline is a licensed pipeline under the Pipelines Act 1967 (NSW) and is subject to operation and maintenance requirements under AS 2885.3:2022: Pipelines: Gas and Liquid Petroleum - Operation and

Maintenance. NSW pipeline licensees are required to maintain pipeline management plans and implement pipeline management systems, with the Pipelines Regulation requiring compliance with relevant AS 2885 requirements.

The pipeline is generally underground and may not be apparent from the surface, except for warning markers along the route. Ampol undertakes regular surveillance to ensure activities such as excavation, construction, vehicle loading or land use changes do not compromise the safe operation and maintenance of the pipeline. A 20m wide existing easement is registered across the site and the indicative pipeline alignment shown below.

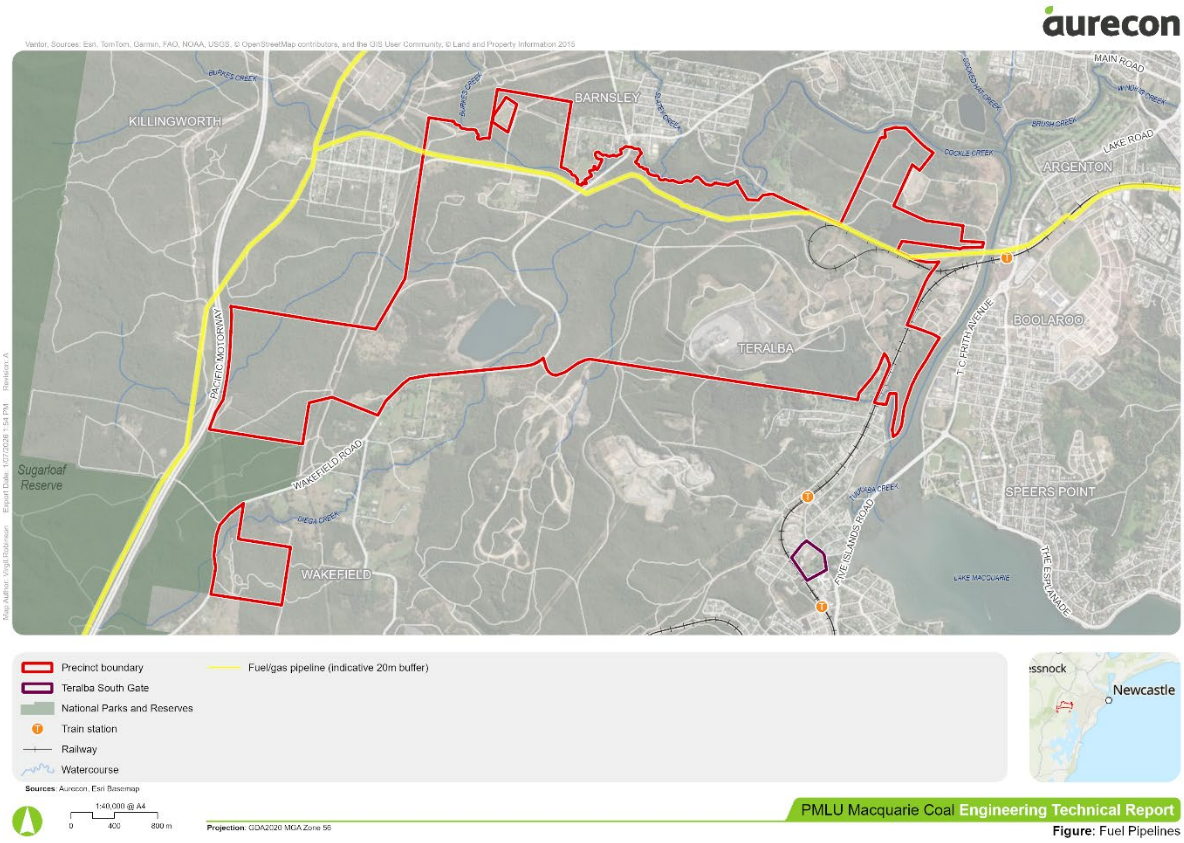

Figure 2-7 AMPOL Fuel Pipeline (Geosciences Australia, 2026)

The pipeline corridor should be treated as a land use and infrastructure constraint. The registered easement width and exact pipeline alignment should be confirmed through title search, BYDA plans, survey and direct liaison with Ampol. BYDA identifies a series of controls that apply to works within or near the pipeline easement. Fundamentally development over or near the pipeline will not be permissible and will be subject to referral to Ampol.

Land use changes, including major residential development or sensitive uses such as schools, childcare centres, hospitals and aged care facilities, may require a formal **Land Use Change Safety Management Study** under AS 2885.6. The measurement length must be confirmed by Ampol for the specific pipeline section. Industrial land uses along the corridor may prove to be a more complementary use for the site. Sufficiently locating road and open space along the corridor will enhance the pipelines protection.

Ampol's BYD response states that written approval must be obtained before any works proceed near the pipeline. Ampol also requires the pipeline to be physically located by Ampol or its representative before works, and an Ampol work permit is required prior to site works.

Opportunities

The presence of the Ampol pipeline does not necessarily preclude development of the broader precinct, but it will influence the Precinct Master Plan, subdivision layout, road hierarchy, service corridors, stormwater design and future staging. The pipeline corridor could potentially be integrated as a linear open space, access, landscape or infrastructure corridor where it remains accessible, unencumbered by structures and designed in accordance with Ampol requirements.

Early coordination with Ampol may also allow the pipeline corridor to be spatially resolved in the structure plan, reducing later uncertainty for developers and infrastructure designers. Future road and service crossings should be rationalised to minimise the number of pipeline interfaces and to ensure any crossings are perpendicular, constructible and capable of being approved.

Constraints

The pipeline is a material constraint to land use planning within the northern portion of the Precinct. Development should avoid placing sensitive or high-occupancy uses within the pipeline measurement length unless a Land Use Change Safety Management Study confirms that the risk profile is acceptable and that any required mitigation measures can be implemented. Sensitive land uses such as schools, childcare, hospitals, aged care and major residential development are likely to require specific assessment if located within the relevant AS 2885.6 measurement length.

Any works with potential vibration impact within 20 m require Ampol supervision and further assessment.

Recommended Activities

The following are recommended as the project progresses.

- Confirm the registered easement, surveyed pipeline alignment, depth of cover and associated title restrictions. A 20m easement is registered on title the exact pipeline location within the easement should be surveyed as DA’s progress.
- Liaise with Ampol to confirm applicable DA planning referral requirements, including the relevant AS 2885.6 measurement length for the pipeline section through the study area. Referral to Ampol will be automatically enacted for any DA’s that approach/cross or intersect the easement or pipeline for Ampol approval.
- Formalise a pipeline constraint overlay showing the easement, 5 m structure exclusion zone, 20 m works / vibration referral zone, 30 m blasting approval zone and any confirmed AS 2885.6 land use change referral area. Zones to be confirmed with Ampol.
- Avoid sensitive or high-occupancy land uses within the confirmed measurement length unless supported by a Land Use Change Safety Management Study.
- Locate buildings, high-value assets, stormwater basins, major earthworks and permanent stockpiles outside the easement and minimum exclusion areas.
- Rationalise roads and utility crossings to minimise interfaces with the pipeline and ensure crossings are perpendicular and capable of meeting Ampol’s separation and approval requirements.
- Include a future development control requiring BYDA enquiry, Ampol approval, work permit, positive pipeline location and supervised works before any excavation, construction, compaction, crane lift, service crossing, landscaping or drainage works near the pipeline.’

2.4.3 Telecommunications

Telstra & NBN

Telecommunications and NBN infrastructure is relatively sparse across the site. Two main connection points to the network exist along Powerhouse Road to the East and from The Broadway to the West (connection the current West End Colliery site head office), where a small exchange is present. Generally there is no major telecommunications infrastructure passing through the site that would present a constraint to development. The minor connections mentioned above would enable unlocking of initial development however additional cabling and network upgrades would be expected as demand grows. Large scale customers would need their own connection and this is not uncommon.

The southern portions of the site along Wakefield Road are connected to the copper Telstra network but no Fibre Optic is present. This Project presents an opportunity to extend the network to the south and would also help support the Wakefield area (including Wakefield School) to the NBN network. Many properties in this southern region connect to NBN's Skymuster Satellite Service.

The Teralba Southgate site has good access to NBN through the existing Telstra network that surrounds the site. Optic Fibre is currently connected to the site.

Mobile & RF Coverage

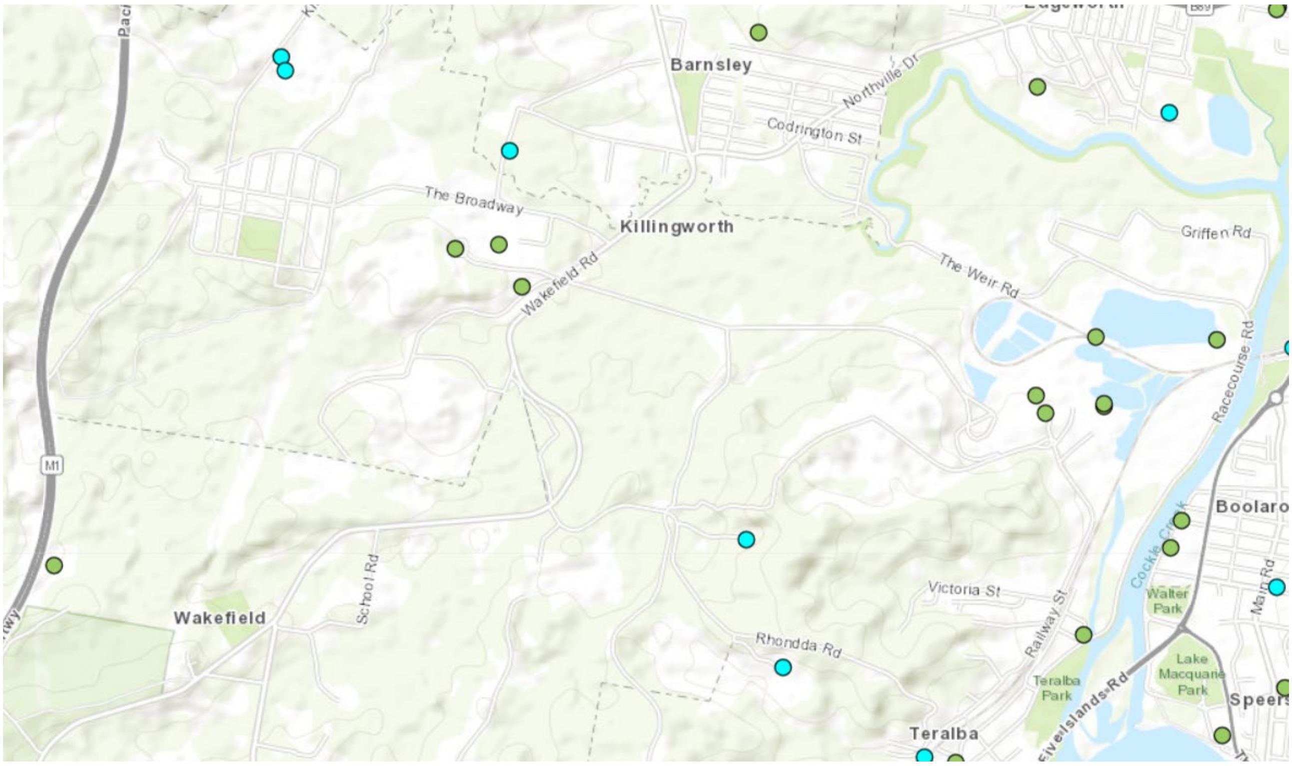

Figure 2-8 ACMA Radio/Telecommunications Towers (ACMA, 2026)

Across the site there are 10 mobile and Radio Frequency communication towers/devices, Figure 2-8 and Table 2-8. The majority of these are expected to be local radio communications related and therefore not significant to operations outside of the site. With the exception of one site **1a The Broadway Killingworth,** all other sites are local in nature or disused and do not limit development potential. The Broadway site operates 2 x 475MHz frequency radio towers and are registered to Oceanic Coal Australia Limited. This site should not impact development potential and may be removed in the future.

Table 2-8 ACMA Radio/Telecommunications Towers (ACMA, 2026)

Name	Assignments	Site_Id
Stockton Borehole Colliery Teralba	0	5824
Oceanic Coal Aust Site West Wallsend Colliery Via KILLINGWORTH	0	5832
Macquarie Coal Prep Plant Stockton Borehole Site Teralba	0	250114
West Wallsend Colliery No 3 Fan Shaft Near KILLINGWORTH	0	250597
Rail Shunting Yard Cockle Creek	0	403903
Westside Mine Site Off Wakefield Rd TERALBA	0	404088

Name	Assignments	Site_Id
Train Operations Stockton Borehole Colliery Teralba	0	404535
Control Tower Coal Prep Plant Stockton Borehole Teralba	0	9014778
Coal Bin West Wallsend Colliery Wakefield Road Killingworth	0	9014779
1a The Broadway Killingworth	**2**	**10033696**

Seeing ‘no assignments’ is common, some reasons include:

- The licensed radios are turned off: The mine's former communication networks (like two-way radios, telemetry, or microwave links) were decommissioned when operations stopped, and their licenses were surrendered.
- The physical tower is likely still standing: The ACMA database keeps the geographic record of the physical mast or structure, even after the gear is removed or powered down.
- It could still be hosting "unlicensed" gear: The old tower might have been repurposed for equipment that doesn't require ACMA registration (like basic point-to-point Wi-Fi links, environmental sensors, or amateur radio).

All sites within the study areas have full coverage and access to 4G and 5G. With future increased load within development additional towers and or strengthening may be required.

3 Precinct Master Plan Assessment

The following sections discuss key findings for utilities servicing based on the current Precinct Master Plan alongside existing desktop information gathered during the baseline analysis.

3.1 Proposed Structure Plan Land Use Assumptions

Land use assumptions for the utilities demand assessments were made in line with the land uses identified for the Precinct Master Plan by GHD in June 2026. The sites within the precinct are displayed in Figure 3-1 and the land use assumptions are shown in Table 3-1.

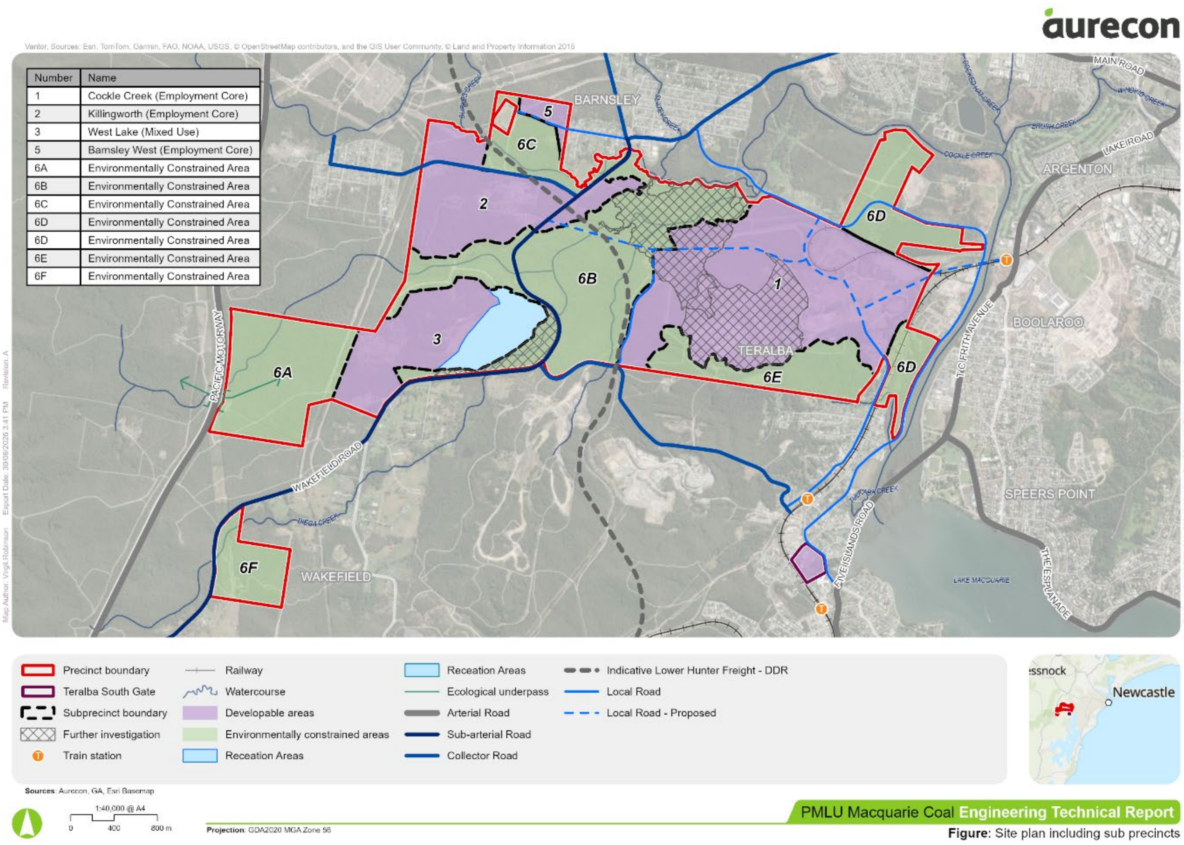

Figure 3-1 Precinct Structure Plan (June 2026)

Table 3-1 Precinct Master Plan Proposed Land Uses

Area	Name	Description	Land Use (Typology)	Land Area (ha)	Assigned Land Use (utilities)
1	Cockle Creek [Employment Core]	Residual developable land	Circular Economy, Waste and Recycling; Industrial and Advanced Manufacturing; Data Centres and Digital Infrastructure; Freight Logistics, Distribution and Supply Chain Hub; Energy Transition & Renewable Infrastructure	168.40	Heavy industrial
2	Killingworth [Employment Core]	Developable land	Circular Economy, Waste and Recycling; Industrial and Advanced Manufacturing; Data Centres and Digital Infrastructure; Freight Logistics, Distribution and Supply Chain Hub; Energy Transition & Renewable Infrastructure	112.20	Light industrial
3	West Lake [Mixed Use]	Developable land	Community, Recreation, Agriculture and Residential	89.40	Commercial
4	Teralba Southgate [Mixed Use]	Developable land	Residential mixed use	7.00	Residential units/flats
5	Barnsley West	Developable land	Light industrial, commercial	11.00	Light Industrial
6	Environmentally Constrained Areas	Non-Developable land	-	-	-

3.2 Water servicing strategy

This section covers the proposed potable water servicing strategy for the Precinct Master Plan. It outlines the demand assessment, the infrastructure required to support development, the recommended staging of works, and the constraints and opportunities that influence the servicing approach.

3.2.1 Demand Assessment

Water demands have been calculated based on the Water Supply Code of Australia, Hunter Water Edition, Version 2 (WSA03-2011). The adopted demand rates are summarised in Table 3-2.

It is noted that;

- the adopted demand values from WSA03-2011 typically provide a conservative assessment of water demand for each land use for the purpose of designing for infrastructure capacity, as developed by Hunter Water.
- the demand assessment may be refined in consultation with Hunter Water as details of the precinct developments are progressed, including details of land use, water efficiency measures, stormwater harvesting and on-site water reuse.
- the land use categories at this stage are based on assumptions which are outlined in Section 3.1. These assumptions are subject to change during later stages of the precinct development.

Table 3-2 Hunter Water Potable Water Supply Design Demands

Category	Average		Peak				Extreme		95th Percentile		
	average annual demand AAD kL	average day demand ADD kL/day	peak day factor PDF	peak day demand PDD kL/day	peak hour factor PHF	peak hour demand PHD kL/day	extreme day factor EDF	extreme day demand EDD kL/day	95th p factor 95F	95th p peak day demand 95PDD kL/day	95th p peak hour demand 95PHD kL/day
Residential House Ncle & Lake Macq LGA	255	0.70	2.25	1.57	2.02	3.18	1.15	1.81	1.8	1.26	2.54
Flats/Units	130	0.36	2.20	0.78	2.02	1.58	1.15	0.90	1.76	0.63	1.27
Industrial – Light	4,200/ha	11.5/ha	1.20	13.8/ha	1.30	18.0/ha	1.15	15.9/ha	1.14	13.1/ha	17.0/ha
Industrial – Medium	9,660/ha	26.5/ha	1.20	31.8/ha	1.30	41.3/ha	1.15	36.5/ha	1.14	30.2/ha	39.2/ha
Industrial – Heavy	25,000/ha	68.5/ha	1.20	82.2/ha	1.30	106.9/ha	1.15	94.5/ha	1.14	78.1/ha	101.6/ha
Commercial	4,200/ha	11.5/ha	1.60	18.4/ha	1.90	35.0/ha	1.15	21.2/ha	1.52	17.5/ha	33.2/ha
Parks And Rural	tbd		1.60		1.50		1.15		1.52		
Unaccounted Water	+ 15%		1.00		1.00		1.00		1.00		

Note:

- Values taken from Table HW 2.4 Water Supply Design Demands in Water Supply Code of Australia, Hunter Water Edition, Version 2 (WSA03-2011)
- Summarised to show only relevant categories used for Macquarie Coal Complex demand assessment

The projected design potable water demand for the Precinct Master Plan, based on Hunter Water's design demands, is outlined in Table 3-3. It is noted that this is a theoretical scenario which is subject to change. If a significant number of high water demand users are added into the Precinct, a modification of the Precinct Master Plan and demand assessment may be required to ensure water demands can be met. It is likely that these high water demand users would be subject to negotiation with Hunter Water for connection into their network.

Table 3-3 Precinct Master Plan Projected Potable Water Demand

Area	Name	Land Use	Equivalent Tenement ET	Average Day Demand ADD kL/d	Average Day Demand + Unaccounted for Water ADD + UFW kL/d	Peak Day Demand PDD kL/d	Peak Hour Demand PHD kL/d	Extreme Day Demand EDD kL/d
4	Teralba Southgate	flats/units	1,260.00	453.60	521.64	1,236.01	2,496.75	1,421.42
Area	**Name**	**Lane Use**	**Net Developable Area**	**Average Day Demand ADD kL/d**	**Average Day Demand + Unaccounted for Water ADD + UFW kL/d**	**Peak Day Demand PDD kL/d**	**Peak Hour Demand PHD kL/d**	**Extreme Day Demand EDD kL/d**
1	Cockle Creek	industrial – heavy	146.12	10,009.40	11,510.81	12,011.28	15,614.67	13,812.98
2	Killingworth	industrial – light	77.42	890.32	1,023.86	1,068.38	1,388.89	1,228.63
3	West Lake	commercial	40.22	462.48	531.86	739.97	1,405.95	850.97
5	Barnsley West	industrial – light	7.97	91.68	105.44	233.89	304.05	268.97
			Totals (all areas)					
				ADD	**ADD + UFW**	**PDD**	**PHD**	**EDD**
			ML/d	11.91	13.69	15.29	21.21	17.58
			kL/d	11,907.48	13,693.61	15,289.54	21,210.31	17,582.97
			L/s	137.82	158.49	176.96	245.49	203.51

High Water Consumers

The demand assessment has been based on Hunter Water's adopted demand rates which do not account for high water consumers. Any developments with significant water usage, such as data centres, will require case-by-case investigations to determine if they can be supported by Hunter Water and the Precinct's potable water network. These assessments should also consider suitable non-potable water sources to reduce demand to the potable water network.

3.2.2 Proposed Infrastructure

The Precinct sits within, or in close proximity to, Hunter Water's Elermore Vale 1 Reservoir Supply Zone within the South Wallsend and West Lake Macquarie DSP Area. This area is serviced by the Grahamstown Water Treatment Plant, which has a total treatment capacity of 246 ML/d and is forecast to have 17.5 ML/d of spare capacity in 2029, see Section 2.1.1. This available capacity would be sufficient to service the entire Precinct and Teralba Southgate under a peak day demand scenario of 15.3 ML/d, but would not quite be able to service an extreme day demand scenario of 17.6 ML/d. Excluding Teralba Southgate from the equation enables the Precinct to be serviced by Grahamstown Water Treatment Plant at an extreme day demand. It is noted that the Precinct is expected to be delivered in stages, meaning that the requirement for these flows will likely come into effect incrementally and there will not be a need to service the entire flow during Stage 1. These initial figures do not consider the addition of other large developments which may be proposed in the area; however, standard population growth is accounted for in Hunter Water's numbers.

Hunter Water's Elermore Vale 1 Reservoir Supply Zone is serviced by two reservoirs which were found to have a combined total of 110 ML of storage, based on desktop investigations. Elermore Vale 1 Reservoirs have a single inlet/outlet DN900 main. For the purpose of understanding high level capacity within the supply zone, it is assumed that the reservoirs are filling for half the day and emptying for the other half at a velocity of 1.1 m/s which gives a volume of approximately 30 ML per day of flow into the network. Based on these high level calculations, it is assumed that this reservoir system has spare capacity to service the development of the Precinct, this assumption is to be confirmed with Hunter Water as the development progresses.

A formal development application to Hunter Water will be required to confirm the proposed servicing arrangements. Potable Water demands will continue to evolve as greater clarity is reached on development types and as the Precinct Master Plan is refined. As planning develops, if Grahamstown Water Treatment Plant has no spare capacity, it is assumed that Hunter Water would plan upgrades to the facility to meet growth needs.

All proposed potable water pipelines will follow the existing road corridors and alignments of proposed roads, where possible, within the Macquarie Coal Complex. The pipelines should be allocated a location within the road corridor, along with the other utility services. It is noted that servicing plans only show trunk infrastructure with the reticulation network to be designed in line with development within the Precinct.

Proposed potable water infrastructure and sizing has been determined built on a base case of 100% of water demand being supplied by a potable water network. No accounts have been made for the use of recycled or raw water on site to supplement the potable water supply. Opportunities to reduce potable water demand, such as the use of raw or recycled water on site, should be further investigated for feasibility in line with development refinement.

Main Precinct Study Area

Due to the large size of the main Precinct and the distribution of the developable areas, the site was split into two distinct sections for the purpose of water servicing strategies. The eastern section of the Precinct to be serviced via an existing crossing from Boolaroo into Area 1 and the western section of the Precinct to be serviced with a new pipeline or reservoir from Edgeworth/Barnsley into Areas 2, 3 and 5.

Eastern Precinct

The eastern section of the main Precinct encompasses Area 1 (Cockle Creek), which includes 299 ha of total land area with approximately 110 ha of this land being designated for further investigation due to the presence of the tailings dam. The developable land has been earmarked for heavy industrial use which has been reflected in the demand assessment.

The eastern Precinct area encompasses the area outlined in Table 3-4.

Table 3-4 Eastern Precinct Associated Areas

Area	Name	Total Land Area (ha)	Developable Area (ha)	Proposed Land Use
1	Cockle Creek	299.18	188.55	Heavy Industrial

This eastern section of the precinct is currently serviced by a Hunter Water DN150 potable water pipeline. The pipeline breaks off from a DN300 pipeline after crossing Cockle Creek, a DN600 also crosses the creek at this location. The pipeline then crosses the northeastern section of Area 1 up to The Weir Road. Due to the scale of development proposed in this section of the Precinct, the existing main is deemed to be undersized to support the future development.

Proposed Servicing

It is proposed to utilise the existing crossing of Cockle Creek to connect Area 1 to a larger pipeline. A connection off the DN600 is assumed to be feasible, however, consultation with Hunter Water is required to determine if there is spare capacity in the network to enable this connection. This proposed main would then cross through the middle of the site and the reticulation network would branch off to service properties. The proposed servicing strategy is shown in Figure 3-2.

An alternative option is the construction of an onsite reservoir which would fill during low periods and require a smaller sized pipeline connection into the existing network.

Proposed Pipe Sizing

Based on the initial demand assessment, during peak hour flow the site is expected to require 180.73 L/s of potable water flow. A 1.1 m/s pipe velocity was assumed to find the required pipe sizing and found the trunk main into Area 1 would need to be a minimum of DN500 to service the site for heavy industrial use. This is an initial estimation and is subject to change with the further development of the Precinct plan. The breakdown of flows and required pipe sizes by area is shown in Table 3-5.

Table 3-5 Eastern Precinct Proposed Potable Water Flows and Pipe Sizing

Area	Peak Hour Flow (L/s)	Proposed Pipe Size
1	180.73	DN500
Total	**180.73**	**DN500**

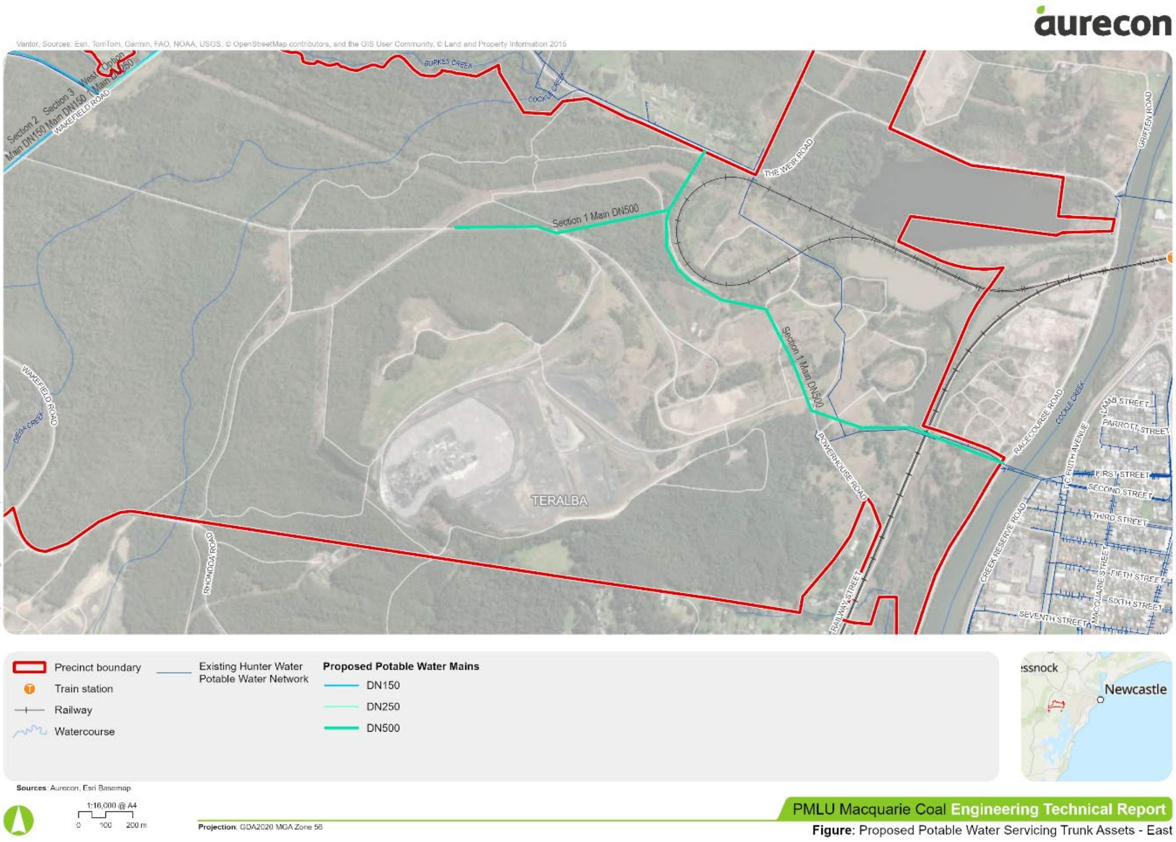

Figure 3-2 Eastern Precinct Proposed Potable Water Servicing Strategy and Pipe Sizing

Western Precinct

The western section of the main Precinct encompasses the below areas outlined in Table 3-6.

Table 3-6 Western Precinct Associated Areas

Area	Name	Total Land Area (ha)	Developable Area (ha)	Proposed Land Use
2	Killingworth	124.18	106.78	Light Industrial
3	West Lake	125.57	89.37	Commercial (tourism / resort)
5	Barnsley West	11.00	11.00	Light Industrial

Area 2 is currently serviced by a Hunter Water DN150 potable water pipeline, which has one offtake within the Precinct to service the existing mine operator office. The pipeline runs down from Barnsley into Killingworth. Due to the scale of development proposed in this section of the Precinct, the existing main is deemed to be undersized to support the future development.

Proposed Servicing

Two servicing options have been proposed for the western section of the Precinct, the preferred option is to be determined in consultation with Hunter Water. Area 5 is proposed to be serviced separately by extending the existing DN100 pipeline along Charlton Street.

Whilst Area 3 is proposed to be an extension off Area 2 for both servicing options, it is located approximately 1.5 km south of Area 2 and the investment to connect this area into the Hunter Water network would be significant. The area has been included in the servicing noting that there is a significant opportunity to service this site as an off grid system to reduce the barrier of connecting it into the wider Hunter Water network.

Option 1

The first option connects into an existing DN500 pipeline in Edgeworth, on Main Road between Corden Street and Garth Street. A new pipeline is then proposed to run within road verges to Area 2 where it would branch out into Area 2 and down to Area 3. It is noted that the spare capacity within this pipeline is unknown and the feasibility of this option is to be determined through consultation with Hunter Water.

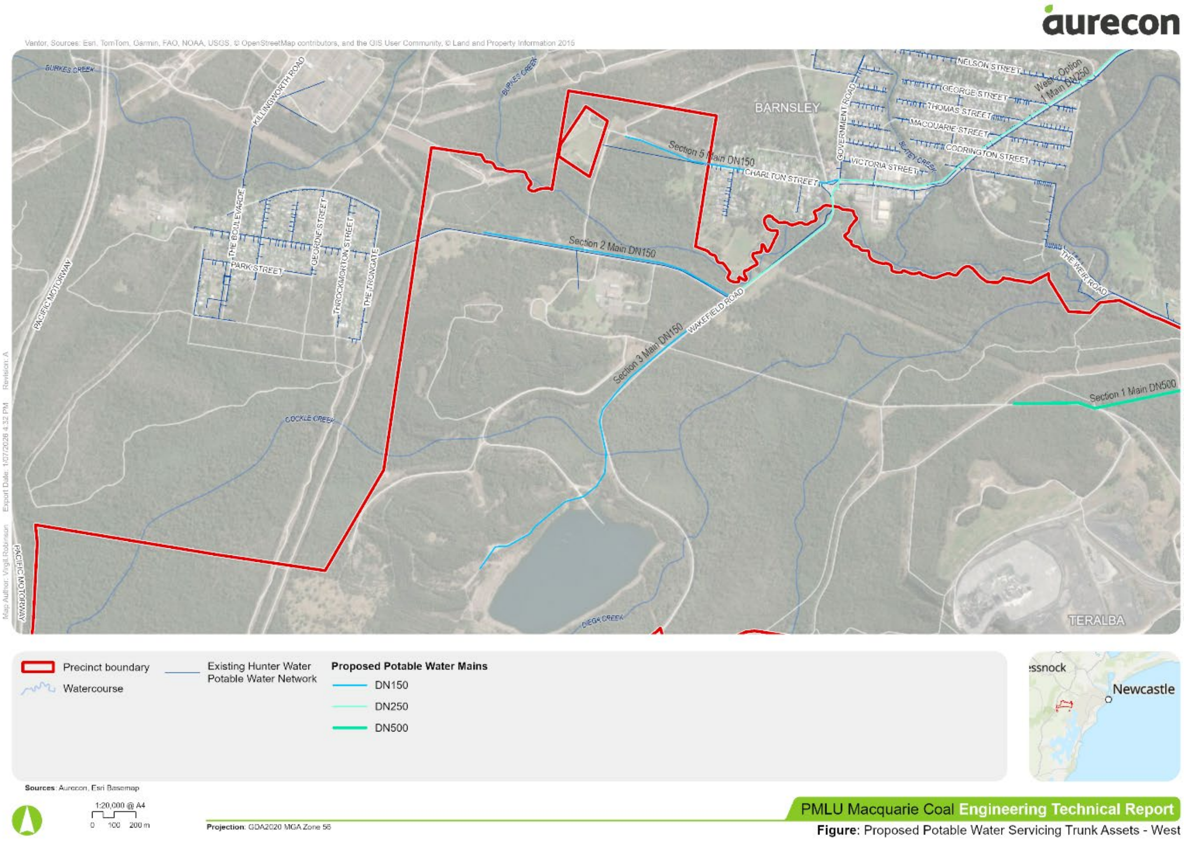

Figure 3-3 Western Precinct Option 1 Proposed Potable Water Servicing Strategy and Pipe Sizing

Option 2

An alternate servicing strategy, Option 2, utilises the existing DN150 pipeline which runs along The Broadway to flow into a new proposed reservoir located within Area 2. The reservoir would be located at a high point and fill during low demand periods within the network. The reticulation network would then be gravity fed into the surrounding areas from the reservoir. Further investigation is required to determine if a booster pump station is required to transport flows from the existing pipeline up to the reservoir. Consultation with Hunter Water is required to determine if the existing pipeline has capacity to service these areas via a new reservoir and potential booster pump station. This option seeks to use a ‘trickle feed’ to constantly fill up a new reservoir. This would allow the existing infrastructure surrounding the site to be used and the difference in Option 1 and 2 then becomes the cost and logistics associated with a new lead-in along Northville Drive vs a new reservoir and pump station within the site. Both options are viable subject to further consultation with Hunter Water.

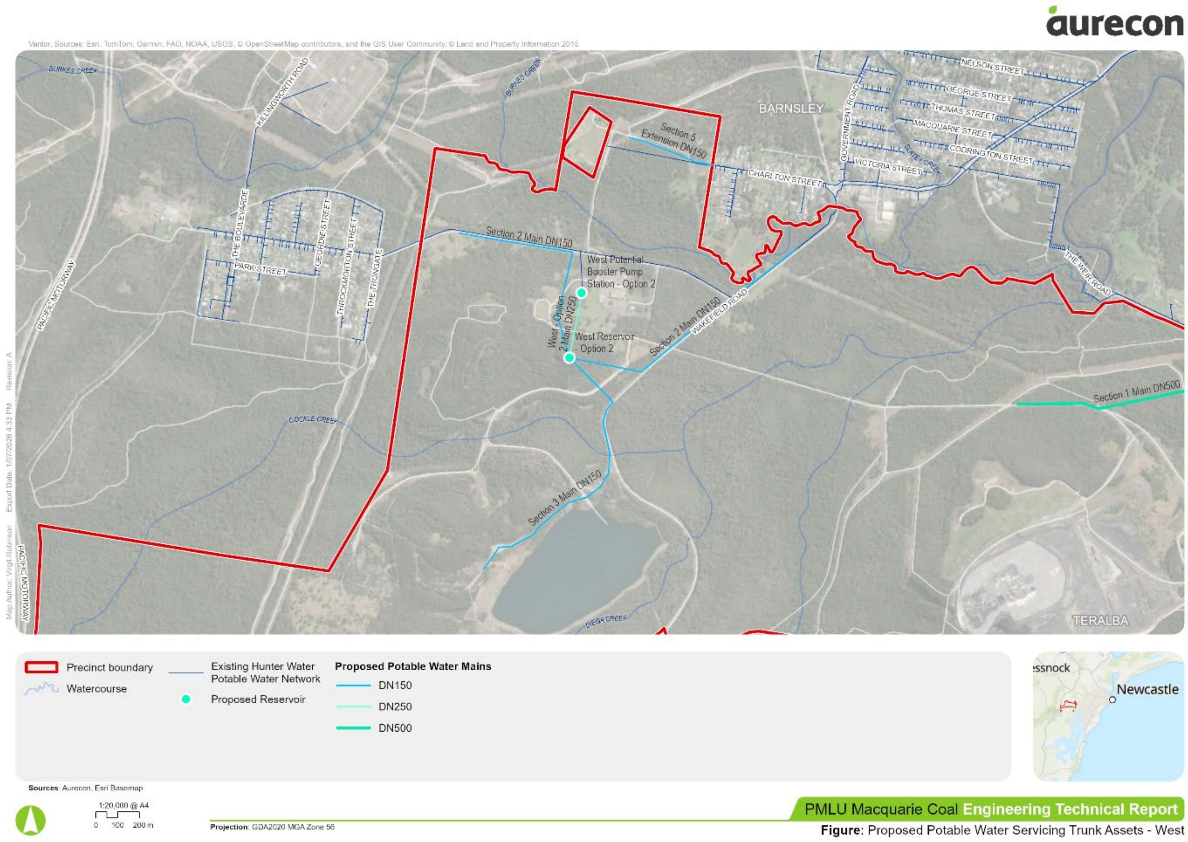

Figure 3-4 Western Precinct Option 2 Proposed Potable Water Servicing Strategy and Pipe Sizing

Proposed Pipe Sizing

From the initial demand assessment, during peak hour flow the combined areas are expected to require 35.87 L/s of potable water flow. A 1.1 m/s pipe velocity was assumed to find the required pipe sizing and found the trunk main into the combined areas to be a minimum of DN250 to service the various land uses. This is an initial estimation and is subject to change with the further development of the Precinct plan. The breakdown of flows and required pipe sizes by area is shown in Table 3-7.

Table 3-7 Western Precinct Proposed Potable Water Flows and Pipe Sizing

Area	Peak Hour Flow (L/s)	Proposed Pipe Size
2	16.08	DN150
3	16.27	DN150
5	3.52	DN150
Total	**35.87**	**DN250**

Teralba Southgate Study Area

Area 4 covers the Teralba Southgate area which includes 7 ha of land on the edge of Teralba and Booragul. The land has been marked as a mixed use residential site and entire site been determined to be developable.

The Teralba Southgate section encompasses the below areas outlined in Table 3-8.

Table 3-8 Teralba Southgate Associated Areas

Area	Name	Total Land Area (ha)	Developable Area (ha)	Proposed Land Use
4	Teralba Southgate	7.00	7.00	Mixed use residential

The site is currently serviced by a Hunter Water potable water pipeline of unknown size which leads off for a DN150 main on York Street. On York Street, there are three other potable water pipelines ranging from DN375 to DN600.

Proposed Servicing

Whilst the existing site is connected into the Hunter Water network, the pipe sizing is unknown and due to the proceeding pipe being a DN150 it is assumed that the pipe will not be able to support development at this location. It is proposed that a new pipeline connect into the site from one of the other pipelines along York Street that better match the required capacity for the site. The exact pipeline is to be determined in line with Hunter Water consultation and better understanding of the capacity within the existing networks. An additional booster pump would be required to service any building with multiple floors, this is typical.

Proposed Pipe Sizing

Based on the initial demand assessment, during peak hour flow the site is expected to require 28.9 L/s of potable water flow. A 1.1 m/s pipe velocity was assumed to find the required pipe sizing and found the connection into the site would need to be a minimum of DN200 to service the site. This is an initial estimation and is subject to change with the further development of the Precinct plan. The breakdown of flows and required pipe sizes by area is shown in Table 3-9.

Table 3-9 Teralba Southgate Proposed Potable Water Flows and Pipe Sizing

Area	Peak Hour Flow (L/s)	Proposed Pipe Size
4	28.90	DN200
Total	**28.90**	**DN200**

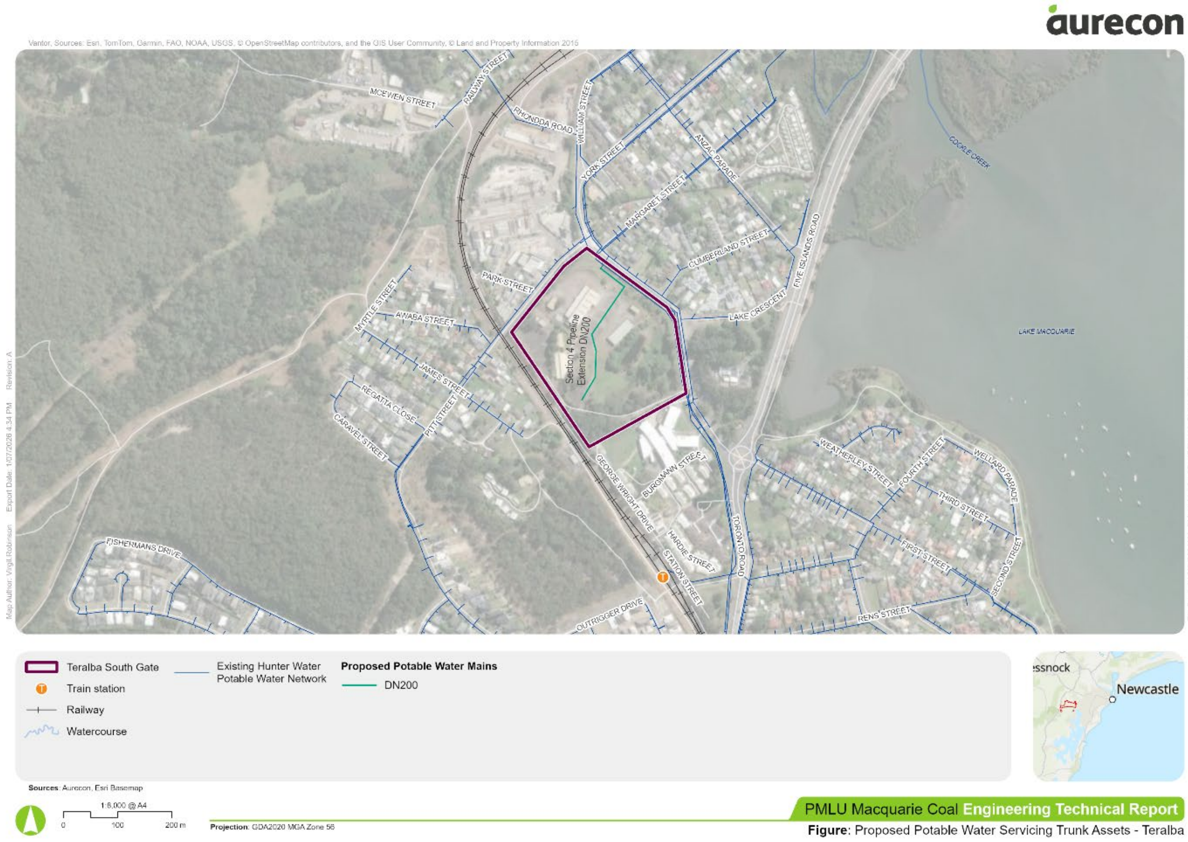

Figure 3-5 Teralba Southgate Proposed Potable Water Servicing Strategy and Pipe Sizing

Wakefield Study Area

The Wakefield site is currently identified as environmentally constrained and therefore does not require a potable water connection. Given its isolated location and the absence of a nearby potable water network, any future need for potable water would likely need to be managed on site. The site does not fall within a Hunter Water supply zone, and the infrastructure required to extend the network to this area would be significant. As such, a substantial and clearly justified water demand would be required to warrant connection to the Hunter Water system.

3.2.3 Infrastructure Staging

For both potable water and wastewater the proposed staging is similar, however, as there is no existing wastewater infrastructure within the precinct, excluding Area 4 at Teralba Southgate, there is a higher barrier of entry to implement wastewater infrastructure.

Proposed infrastructure staging for potable water is shown in Figure 3-6.

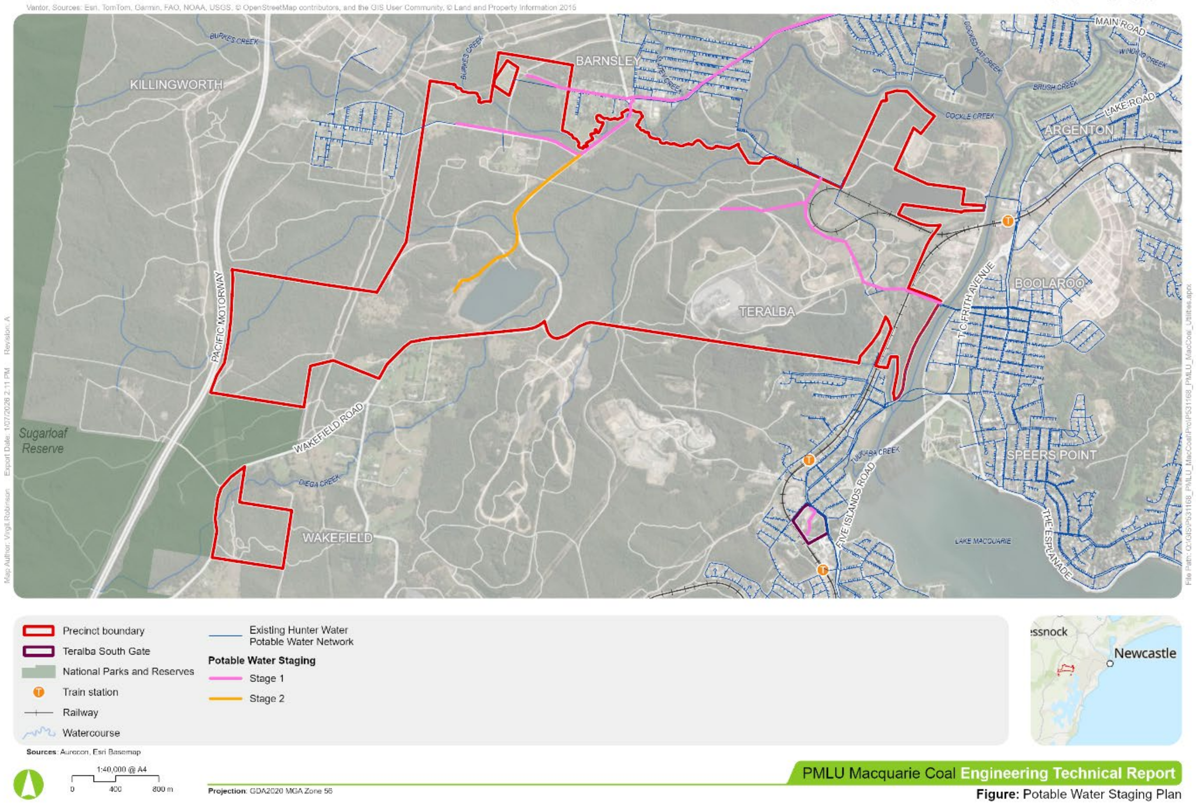

Figure 3-6 Proposed Potable Water Infrastructure Staging

Stage 1

The areas with or near existing potable water infrastructure are earmarked for earlier servicing due to the lower barriers required to supply these sites. This includes; the northern sections of the Precinct, Areas 1 and 2, and Area 4 in Teralba Southgate.

Stage 2

Area 3 is advised to be developed later due to its isolated position and thus the high investment required to service the area. The development of this site has the opportunity to happen sooner if the site is serviced by a decentralised, off-grid system, this will reduce the high investment barrier if the development has low demand.

The servicing of Area 5 is also suggested to be a Stage 2 investment due to it being a standalone system and the investigations required to determine if it can link into the existing network.

3.2.4 SWOT Analysis

The strengths, weaknesses, opportunities and threats to the water servicing of the proposed Precinct Master Plan are outlined in Table 3-10.

Table 3-10 Precinct Master Plan Water Servicing SWOT Analysis

Strengths	Weaknesses
▪ Day 1 access to existing Hunter Water assets within or near Areas 1, 2, 4 and 5 ▪ Numerous potable water pipelines, with varying sizes, in close proximity to Area 5 ▪ Spare capacity within Grahamstown Water Treatment Plant ▪ Good access to alternative water sources such as dams and potential for recycled water reuse from Edgeworth WWTW	▪ Existing pipelines are generally unable to service the ultimate development ▪ No potable water network in or near Area 3 along Wakefield Road ▪ Lead in works along Northville Road would be challenging ▪ Additional booster pumps may be required to reach higher elevations ▪ No existing established treated alternative water supply
Threats	**Opportunities**
▪ Unknown capacity within Hunter Water's potable water networks ▪ Very high water users such as Data Centres	▪ Utilising raw/recycled water, from existing dams and/or Edgeworth WWTW, to supplement the potable water supply and reduce demands on the Hunter Water network. Scheme could also be utilised to benefit the surrounding region. ▪ Effluent flow from Edgeworth Treatment Plant already utilised at Waratah Golf Course ▪ Regional stormwater harvesting scheme to reduce stormwater runoff impacts to local water bodies with the potential to treat water for reuse ▪ Rainwater harvesting for individual properties to reduce their reliance on Hunter Water's potable water network where feasible ▪ Set up Area 3 as an off grid site to reduce the infrastructure barrier to the site ▪ Extend Area 3 connection into Wakefield to service residential properties

3.2.5 Planning and Approval Framework and Control Recommendations

Macquarie Coal Complex will require approvals from Hunter Water in line with the standard Developer Servicing Strategy process. This process is outlined on Hunter Water's website and is subject to change.

It is recommended that the following controls are applied to developments within the Precinct to enhance the efficiency and sustainability of the Macquarie Coal Complex:

- On-site rainwater harvesting for non-potable water usage
- Utilising nearby recycled and raw water sources for non-potable water usage
- *IoT smart water meters*

3.3 Wastewater Servicing Strategy

This section covers the proposed wastewater servicing strategy for the Precinct Master Plan. It outlines the demand assessment, the infrastructure required to support development, the recommended staging of works, and the constraints and opportunities that influence the servicing approach.

3.3.1 Load Assessment

Wastewater discharge loads have been calculated based on the Gravity Sewerage Code of Australia, Hunter Water Edition, Version 2 (WSA 02—2014-3.1). The adopted load rates are summarised in Table 3-11.

It is noted that;

- the adopted load values from WSA02-2014 typically provide a conservative assessment of wastewater discharge for each land use for the purpose of designing for infrastructure capacity, as developed by Hunter Water.
- the load assessment may be refined in consultation with Hunter Water as details of the precinct developments are progressed, including details of land use, water efficiency measures, stormwater harvesting and on-site water reuse.
- the land use categories at this stage are based on assumptions which are laid out in Section 3.1. These assumptions are subject to change during later stages of the Precinct development.

Table 3-11 Hunter Water Wastewater Equivalent Tenements and Storm Allowance

Classification	Module	Unit adopted	ET/ unit	Storm Allowance
Residential (Unit Basis)	Single Cottage	Tenement	1	0.058 / Unit
Residential (Unit Basis)	Flat, Home Unit, Town House (Individual *allotment* thereof)	Flat	2/3	0.058 / Module
Commercial (ii) Area Basis	High Density Zone	Built-up Ha (floor area for multi-storey)	10	0.58 / Net Ha
Industrial (Area Basis)	Multi Purpose Future use unknown	Gross Ha	30-50	0.58 / Unit

Note:

- Values taken from Table N.1 Equivalent Tenements and Storm Allowance in Gravity Sewerage Code of Australia, Hunter Water Edition, Version 2.1 (WSA03:2014-3.1)
- Summarised to show only relevant classifications for Macquarie Coal Complex

The projected design wastewater load for the Precinct Master Plan, based on Hunter Water's design loads, is outlined in Table 3-12. It is noted that this is a theoretical scenario which is subject to change. If a significant number of high wastewater producers are added into the Precinct, a modification of the Precinct Master Plan and load assessment may be required to ensure wastewater loads can be met. It is likely that these high wastewater producers would be subject to negotiation with Hunter Water for connection into their network.

Table 3-12 Precinct Master Plan Projected Wastewater Demand

Area	Name	Land Use	Equivalent Tenement ET	Average Dry Weather Flow ADWF L/s	Peak Dry Weather Flow PDWF L/s	Storm Allowance 50% Reduction SA L/s	Peak Wet Weather Flow PDWF L/s
1	Cockle Creek	industrial – heavy	9,427.27	8,955.91	184.85	54.68	239.53
2	Killingworth	industrial – light	3,203.53	3,043.36	70.08	30.97	101.04
3	West Lake	commercial	160.86	152.82	5.33	4.67	9.99
4	Teralba Southgate	flats/units	840.00	798.00	21.72	0.00	21.72
5	Barnsley West	industrial – light	329.89	313.40	9.76	95.67	105.43
			Totals (all areas)				
				ADWF L/s	PDWF L/s	SA L/s	PDWF L/s
			ML/d	13.26	25.21	16.07	41.27
			kL/d	13263.48	25206.25	16068.69	41274.93
			L/s	12152.08	291.74	185.98	477.72

Note:

- Flows calculated using 0.011 L/s/ET
- Storm allowance reduced by 50% in line with Hunter Water Pressure Sewer Systems Hydraulic Design Guideline (December 2022)

3.3.2 Proposed Infrastructure

Current wastewater connectivity into the Hunter Water network is limited to the Teralba Southgate area, with the broader Precinct being unconnected but located within close proximity to Hunter Water's existing network. Two wastewater treatment plants service the broader Precinct, the Teralba Southgate area feeds into Toronto WWTW and the Precinct is surrounded by sewer system areas which flow into Edgeworth WWTW. As such the Precinct has been divided into three distinct areas for proposed wastewater servicing, these areas will have common routes to take wastewater to the associated wastewater treatment plants.

A formal development application to Hunter Water will be required to confirm the proposed servicing arrangements. Wastewater demands will continue to evolve as greater clarity is reached on development types and as the Precinct Master Plan is refined.

All proposed rising mains will follow the existing road and alignments of proposed roads, where possible. The gravity reticulation pipeline network is to be determined in line with development, alignments within road corridors will be prioritised but it is noted that this configuration is not always feasible, and the pipelines may be required to pass through developable land to meet flow requirements. The pipelines should be allocated a location within the road corridor, along with the other utility services. It is noted that servicing plans only show trunk infrastructure with the reticulation network to be designed in line with development within the Precinct.

The Precinct Study Area

The Precinct is not currently serviced by Hunter Water's wastewater network and as such does not fall within an existing sewer system area. To the north, east and west of the site, wastewater flows are pumped to Edgeworth Wastewater Treatment Works. Edgeworth WWTW is proposed to receive wastewater flows from the Precinct, the plant is much larger and closer to the site than Toronto WWTW; which services the Teralba Southgate area .

Edgeworth Wastewater Treatment Works has a current treatment capacity of 70,000 EP (14ML/d), it is expected to hit this capacity by 2029. Upgrades are planned for the site, but timing is unknown and there is no allocated funding at this stage. As Edgeworth WWTW is nearing capacity, it is assumed that Hunter Water will plan upgrades to the facility to meet growth needs in the region.

Due to the large size of the Precinct and the distribution of the developable areas, the site was split into two distinct sections for the purpose of wastewater servicing strategies. Both sides of the site are proposed to be serviced by a series of pump stations feeding into one major pump station which will carry flows, via a rising main, to Edgeworth WWTW. This configuration is in keeping with the standard strategy which has been utilised in the surrounding suburbs.

Eastern Precinct

The eastern section of the Precinct encompasses Area 1 (Cockle Creek), which includes 299 ha of total land area with approximately 110 ha of this land being designated for further investigation due to the presence of the tailings dam. The developable land has been earmarked for heavy industrial use which has been reflected in the demand assessment.

The eastern section of the main precinct encompasses the below areas outlined in Table 3-13.

Table 3-13 Eastern Precinct Associated Areas

Area	Name	Total Land Area (ha)	Developable Area (ha)	Proposed Land Use
1	Cockle Creek	299.18	188.55	Heavy Industrial

Proposed Servicing

It is proposed to service Area 1 via a series of four pump stations and associated rising mains. These will service four catchment areas, and the rising mains will take flows to a major pump station to transport flows across Cockle Creek to Edgeworth WWTW, see Figure 3-7.

The major pump station for the east is proposed be located at the northern boundary of the site, adjacent to The Weir Road. A rising main is then proposed to run along The Weir Road and then turn to the north and run parallel to the electrical easement up to Cockle Creek. All pump stations are located at low points with the reticulation network proposed to take gravity flows to these locations.

Proposed Pipe Sizing

Based on the initial demand assessment, during peak dry weather flow the site is expected to produce 184.85 L/s of wastewater flow. A flow rate of two and a half times the PDWF and a pipe velocity of 1.5 m/s, were used to find the required pipe sizing and found the main rising main from the site would need to be a minimum of DN600 to service the site. This is an initial estimation and is subject to change with the further development of the Precinct plan. The breakdown of flows and required pipe sizes by catchment areas are shown in Table 3-14.

Table 3-14 Eastern Precinct Proposed Wastewater Flows and Pipe Sizing

Catchment Zone	% of Area	PDWF x 2.5 (L/s)	Proposed Pipe Size to Service Catchment
East Major 1	22%	100.82	DN300
East Minor 1	49%	227.58	DN450
East Minor 2	23%	105.13	DN300
East Minor 3	6%	28.60	DN150
Total (Area 1)		**462.13**	**DN600**

Due to some sewer pump stations servicing multiple catchments, the associated rising mains need to be sized to accommodate the combined flow from these catchments. The resulting pipe sizing for the proposed infrastructure is shown in Table 3-15.

Table 3-15 Eastern Precinct Proposed Rising Main Pipe Sizing

Sewer Pump Station	Catchments Serviced	Total PDWF x 2.5 (L/s)	Proposed Pipe Size to Service Combined Catchments
East Major 1	East Major 1 East Minor 1 East Minor 2 East Minor 3	462.13	DN600
East Minor 1	East Minor 1	227.58	DN450
East Minor 2	East Minor 2 East Minor 3	133.74	DN300
East Minor 3	East Minor 3	28.60	DN150

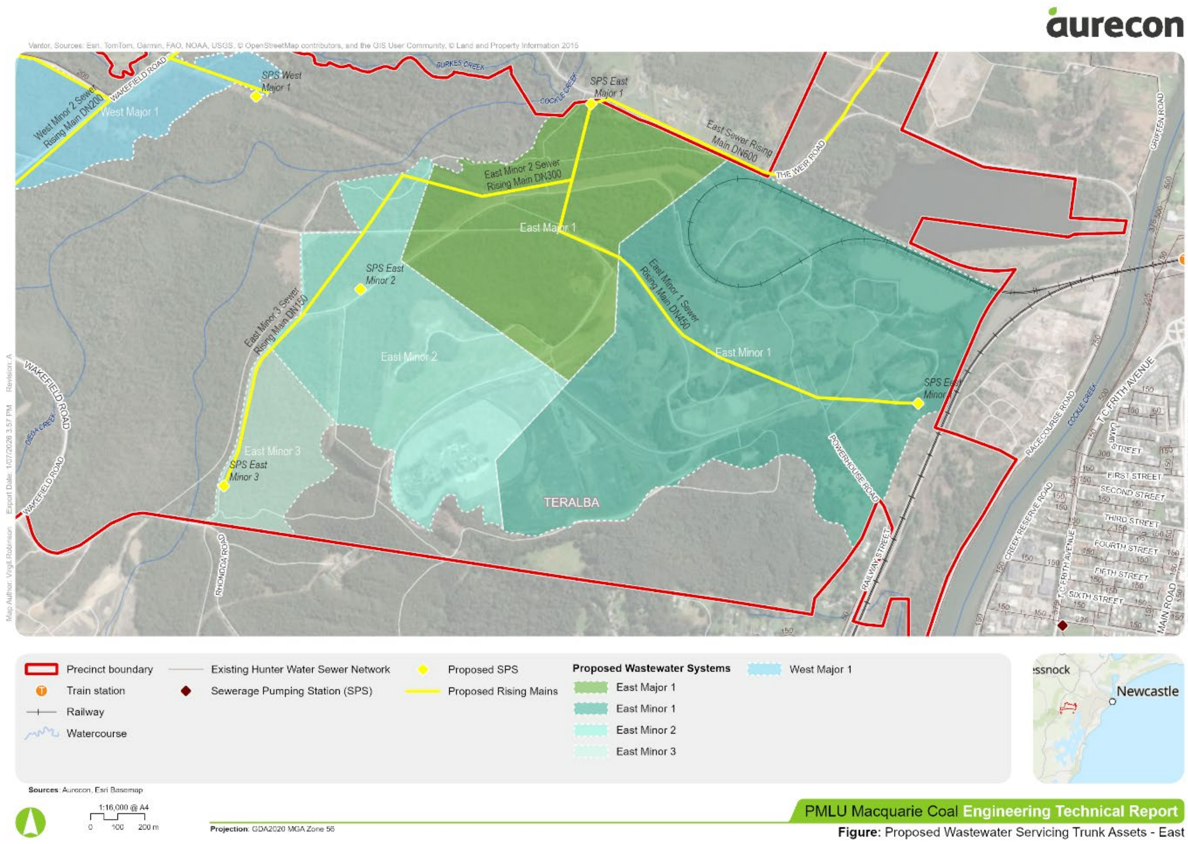

Figure 3-7 Eastern Precinct Proposed Wastewater Servicing Strategy and Pipe Sizing

Western Precinct

The western section of the Precinct encompasses the below areas in Table 3-16.

Table 3-16 Western Precinct Associated Areas

Area	Name	Total Land Area (ha)	Developable Area (ha)	Proposed Land Use
2	Killingworth	124.18	106.78	Light Industrial
3	West Lake	125.57	89.37	Commercial (tourism / resort)
5	Barnsley West	11.00	11.00	Light Industrial

Whilst there is an existing rising main that crosses Area 2 along The Broadway and Wakefield Road, between Killingworth and Barnsley, this main is unable to be tapped into due to the pumped, pressurised nature of the pipeline. As the reticulation networks are confined to the nearby suburbs of Killingworth, Barnsley and Edgeworth, there is no reticulation network that falls within this area of the Precinct.

There are multiple sewer pump stations near the Precinct, however, these have been assumed to have no spare capacity for any additional connections due to the alignment of the Holmesville Rising Main. The Holmesville Rising Main was constructed in 1997, post the construction of the Edgeworth 1 Rising Main (1970), Barnsley 1 Rising Main (1982), and Barnsley 2 Rising Main (1983). Holmesville is located to the north of Barnsley and the rising main passes through Barnsley and Edgeworth, adjacent to the three aforementioned pumping stations, but does not connect into any of these networks. As such, it is assumed that the Barnsley and Edgeworth Systems (Barnsley 1 and 2, Edgeworth 1) are at capacity and additional flow cannot be added into these networks; Killingworth (Killingworth 1 and 2) also flows into these systems so cannot be utilised. Therefore, a new SPS and rising main will be required to directly transfer wastewater to Edgeworth WWTP.

Proposed Servicing

Area 5 is proposed to be serviced separately by connecting into the gravity network at Charlton Street and Bendigo Street.

Areas 2 and 3 are proposed to be serviced via a series of five pump stations and associated rising mains. These will service five catchment areas, and the rising mains will take flows to a major pump station to transport flows across Cockle Creek to Edgeworth WWTW, see Figure 3-8.

The major pump station in the west is proposed to be located on the western boarder of Area 2, near the electrical easement off Wakefield Road. The associated rising main will run from there to Edgeworth WWTW following a similar alignment to the existing rising mains from Killingworth and Barnsley. All pump stations are located at low points and with the reticulation network proposed to take gravity flows to these locations.

Whilst Area 3 is proposed to be an extension off Area 2, it is located approximately 1.5 km south of Area 2 and the investment to connect this area into the Hunter Water network would be significant. The area has been included in the servicing noting that there is a significant opportunity to service this site as an off grid system to reduce the barrier of connecting it into the wider Hunter Water network.

Proposed Pipe Sizing

Based on the initial demand assessment, during peak dry weather flow Areas 2 and 3 are expected to produce a combined 75.41 L/s of wastewater flow. A flow rate of two and a half times the PDWF and a pipe velocity of 1.5 m/s, were used to find the required pipe sizing and found the rising main from the area would need to be a minimum of DN375 to service the site. Area 5 was found to require a DN150 pipeline based on the 9.76 L/s of expected peak dry weather flow. This is an initial estimation and is subject to change with the further development of the Precinct plan.

The breakdown of flows and required pipe sizes by catchment areas are shown in Table 3-17.

Table 3-17 Western Precinct Proposed Wastewater Flows and Pipe Sizing

Catchment Zone	% of Area	PDWF x 2.5 (L/s)	Proposed Pipe Size to Service Catchment
West Major 1	29%	51.18	DN200
West Minor 1	52%	91.84	DN250
West Minor 2	18%	32.17	DN150
Total Area 2		175.19	DN375
West Minor 3	81%	10.78	DN150
West Minor 4	19%	2.54	DN150
Total Area 3		13.32	DN150
Area 5	100%	24.41	DN150
Total (all)		**212.92**	**DN400**

Due to some sewer pump stations servicing multiple catchments, the associated rising mains need to be sized to accommodate the combined flow from these catchments. The resulting pipe sizing for the proposed infrastructure is shown in Table 3-18.

Table 3-18 Western Precinct Proposed Rising Main Pipe Sizing

Sewer Pump Station	Catchments Serviced	Total PDWF x 2.5 (L/s)	Proposed Pipe Size to Service Combined Catchments
Area 5	Area 5	24.41	DN150

Sewer Pump Station	Catchments Serviced	Total PDWF x 2.5 (L/s)	Proposed Pipe Size to Service Combined Catchments
West Major 1	West Major 1 West Minor 1 West Minor 2 West Minor 3 West Minor 4	188.51	DN375
West Minor 1	West Minor 1	91.84	DN250
West Minor 2	West Minor 2 West Minor 3 West Minor 4	45.50	DN200
West Minor 3	West Minor 3 West Minor 4	13.32	DN150
West Minor 4	West Minor 4	2.54	DN150

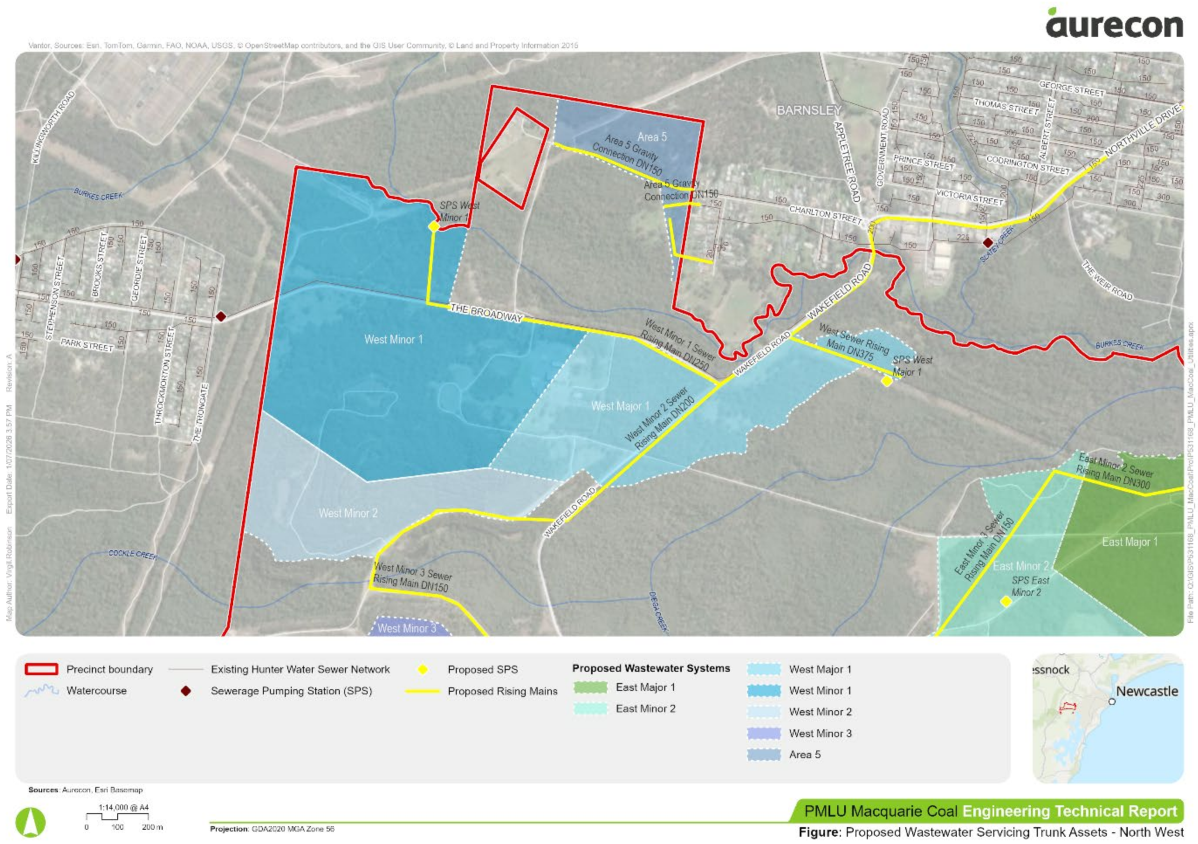

Figure 3-8 North Western Precinct Proposed Wastewater Servicing Strategy and Pipe Sizing

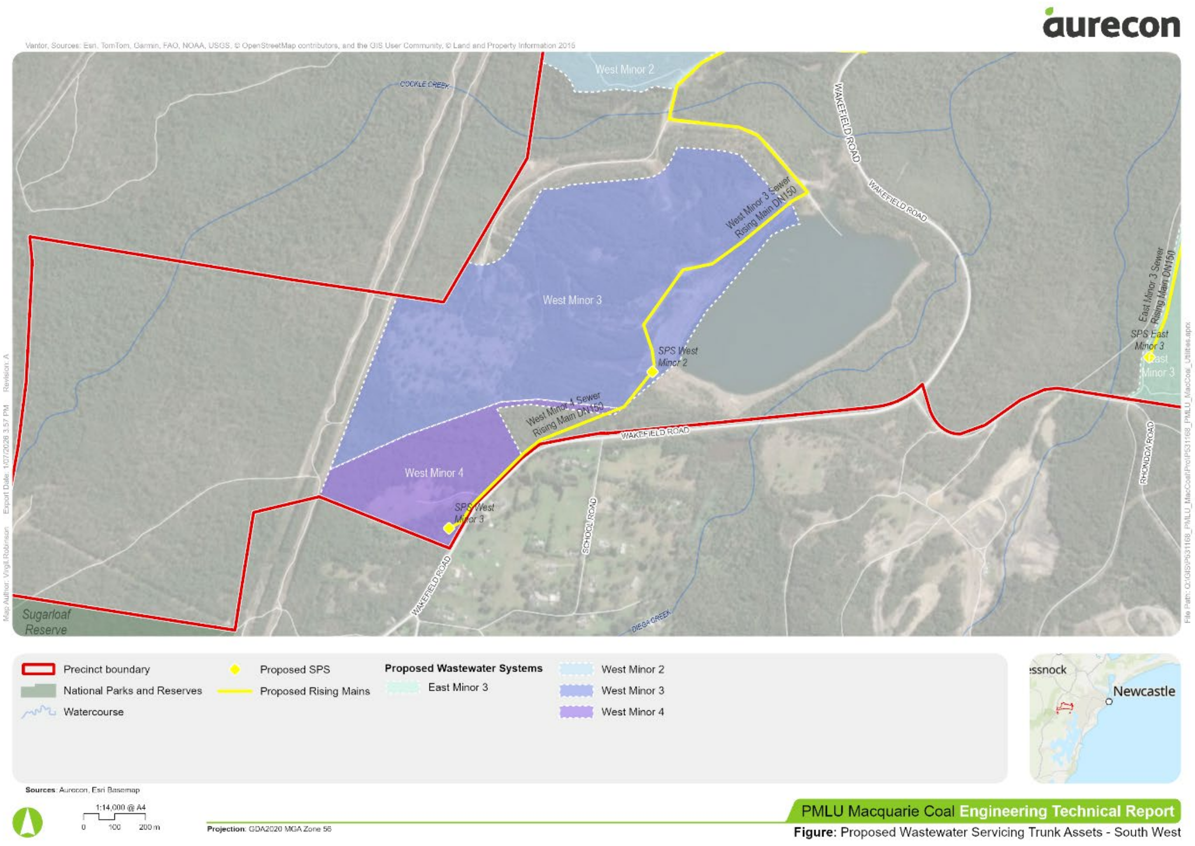

Figure 3-9 South Western Precinct Proposed Wastewater Servicing Strategy and Pipe Sizing

Teralba Southgate Study Area

Area 4 covers the Teralba Southgate area which includes 7 ha of land on the edge of Teralba and Booragul. The land has been marked as a mixed use residential site and entire site been determined to be developable.

The Teralba Southgate section encompasses the below areas outlined in Table 3-19.

Table 3-19 Teralba Southgate Associated Areas

Area	Name	Total Land Area (ha)	Developable Area (ha)	Proposed Land Use
4	Teralba Southgate	7	7	Mixed use residential

Area 4 sits within Hunter Water's Booragul 1 Sewer System Area, which connects to Toronto Wastewater Treatment Works via Booragul 1 Sewer Pump Station.

Toronto Wastewater Treatment Works has a current treatment capacity of 42,000 EP (8.4ML/d), it is expected to have some capacity beyond 2034. In 2029, the treatment plant is expected to have 0.4 ML/d of spare capacity which is sufficient to service the proposed development at Teralba Southgate.

The site is currently serviced by a DN150 Hunter Water wastewater pipeline which runs along the back of residential properties then crossed York Street into the site. A larger pipeline, DN400, flows down to Booragul 1 from nearby Lake Crescent. There is a DN750 running around the perimeter of the site, along Pitt Street and the railway corridor, this is an effluent main from Edgeworth WWTW which cannot be connected into.

Proposed Servicing

Based on the load assessment, the flows produced on the site are likely to require a DN200 pipe to service the site. Due to the existing connection being DN150, this pipeline is expected to be unable to service the entire Teralba Southgate site. It is proposed to alter the connection point into the network to tap into the DN400 pipeline which runs from Lake Crescent down to the pump station, Figure 3-10. Based on initial investigations, a gravity pipeline to this tie in point appears feasible. The exact pipeline and alignment are to be determined in line with Hunter Water consultation and better understanding of the capacity within the existing networks.

Proposed Pipe Sizing

Based on the initial demand assessment, during peak dry weather flow the site is expected to produce 21.72 L/s of wastewater flow. A flow rate of two and a half times the PDWF and a pipe velocity of 1.5 m/s, were used to find the required pipe sizing and found the gravity main from the site would need to be a minimum of DN200 to service the site. This is an initial estimation and is subject to change with the further development of the Precinct plan. The breakdown of flows and required pipe sizes by catchment areas are shown in Table 3-20.

Table 3-20 Teralba Southgate Proposed Wastewater Flows and Pipe Sizing

Catchment Zone	% of Area	PDWF x 2.5 (L/s)	Proposed Pipe Size
Area 4	100%	54.29	DN200
Total		**54.29**	**DN200**

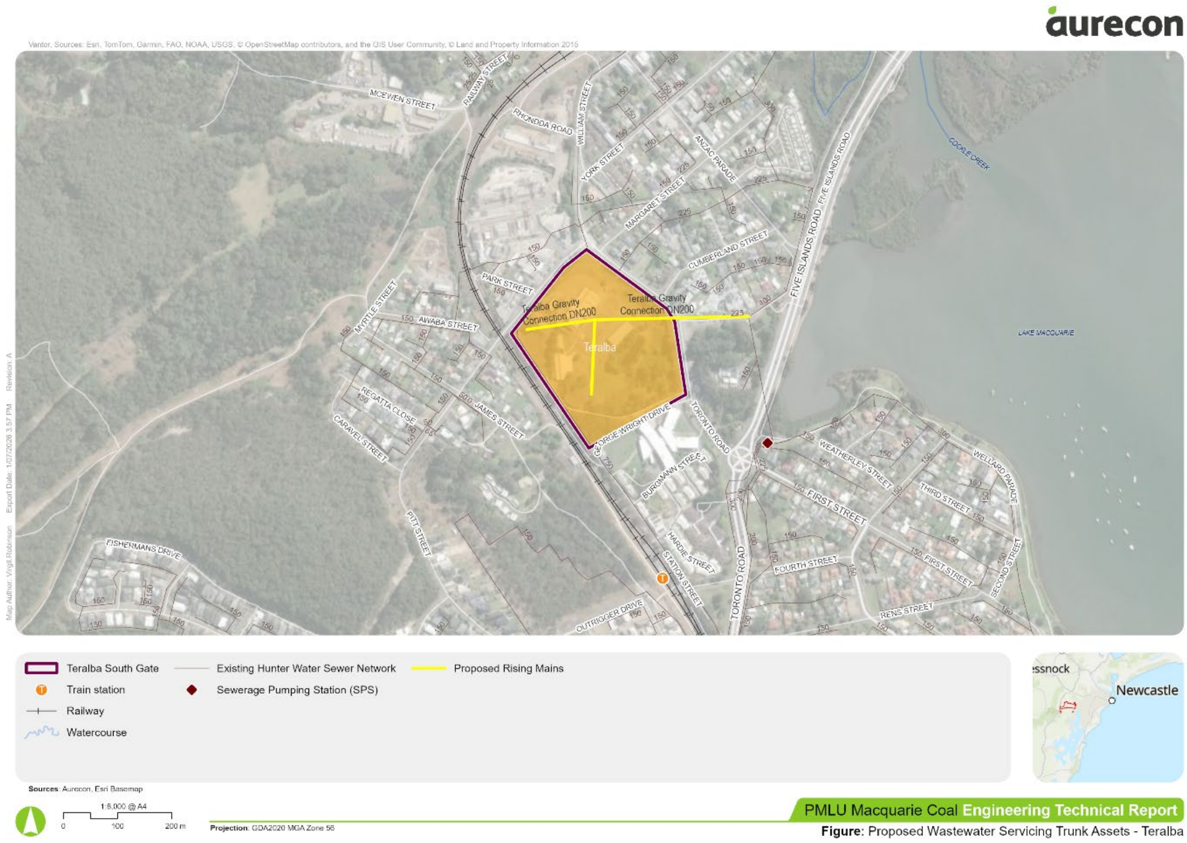

Figure 3-10 Teralba Southgate Proposed Wastewater Servicing Strategy and Pipe Sizing

Wakefield Study Area

The Wakefield site is currently identified as environmentally constrained and therefore does not require a wastewater connection. Given its isolated location and the absence of nearby infrastructure, any future wastewater management would likely need to occur on site. Since the area falls outside Hunter Water's service network, extending infrastructure into this area would require substantial investment. Consequently, a significant wastewater demand would be necessary to justify connection to the Hunter Water system.

Precinct Wide Wastewater Servicing Strategy

The overall servicing strategy for the Precinct is shown in Figure 3-11.

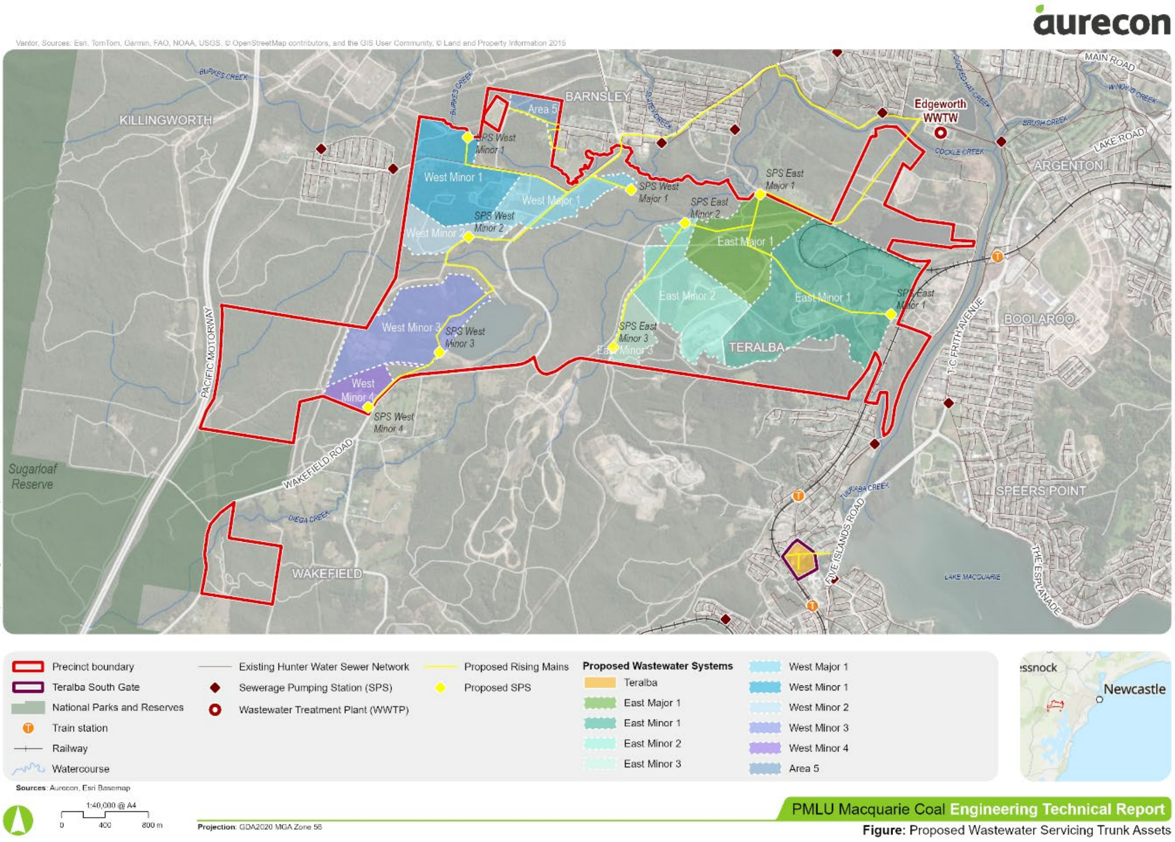

Figure 3-11 Proposed Wastewater Servicing Strategy

3.3.3 Infrastructure Staging

For both potable water and wastewater the proposed staging is similar, however, as there is no wastewater infrastructure within the Precinct, there is a higher barrier of entry to implement wastewater infrastructure.

Proposed infrastructure staging for potable water is shown in Figure 3-12.

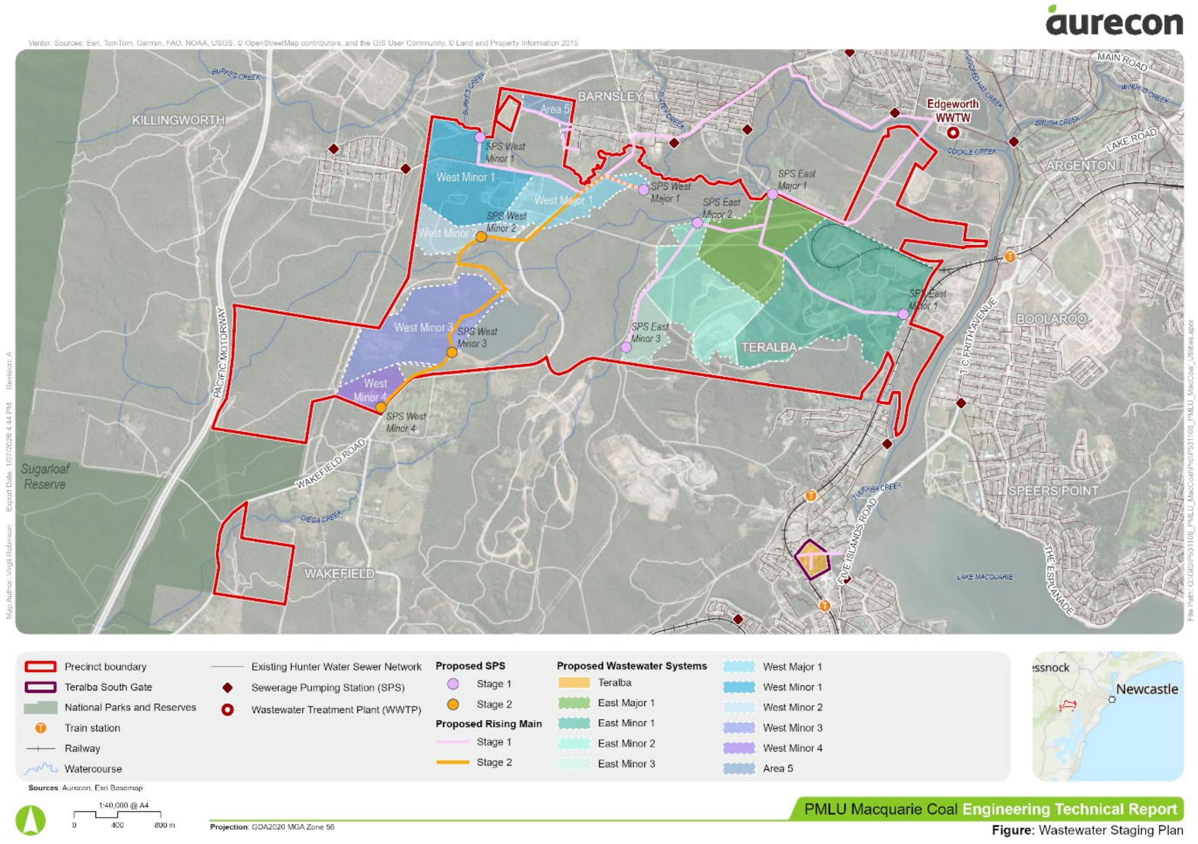

Figure 3-12 Proposed Wastewater Infrastructure Staging

Stage 1

The areas with or near existing wastewater infrastructure are earmarked for earlier servicing due to the lower barriers required to supply these sites. This includes; the northern sections of the Precinct, Areas 1 and 2, and Area 4 in Teralba Southgate.

Stage 2

Area 3 is advised to be developed later due to its isolated position and thus the high investment required to service the area. The servicing of Area 5 is also suggested to be a Stage 2 investment due to it being a standalone system and the investigations required to determine if it can link into the existing network.

3.3.4 SWOT Analysis

The strengths, weaknesses, opportunities and threats to service the wastewater discharge of the proposed Precinct Master Plan are outlined in Table 3-21.

Table 3-21 Precinct Master Plan Wastewater Servicing SWOT Analysis

Strengths	Weaknesses
▪ Teralba Southgate has an existing wastewater connection and in close proximity to larger pipelines ▪ Enough capacity in Toronto WWTW to service Teralba Southgate development in 2029 ▪ Proximity of the Precinct to Edgeworth WWTW	▪ No existing wastewater network within the Precinct ▪ No spare capacity in Edgeworth WWTW by 2029 ▪ No spare capacity in Killingworth, Barnsley and Edgeworth wastewater network systems to take on addition flow from the site

Threats	Opportunities
▪ Required trenchless crossings of Cockle Creek to connect into Edgeworth WWTW	▪ Extend Area 3 connection into Wakefield to service residential properties ▪ Set up Area 3 as an off grid site to reduce the infrastructure barrier to the site

3.3.5 Planning and Approval Framework and Control Recommendations

Macquarie Coal Complex will require approvals from Hunter Water in line with the standard Developer Servicing Strategy process. This process is outlined on Hunter Water's website and is subject to change.

3.4 Electricity Servicing Strategy

The Electricity Servicing Strategy, as summarised in this section, outlines the impacts to electricity supply infrastructure, and sets out what is needed to ensure the timely delivery of key electricity infrastructure upgrades for the Precinct and Teralba Southgate, while ensuring existing customers in the surround areas continue to be serviced as well.

Electricity supply infrastructure will be designed and constructed in accordance with Ausgrid and Transgrid's technical, planning and regulatory requirements. The proposed Precinct is located adjacent to the Argenton 132/33/11 kV Zone Substation (ZS), owned and operated by Ausgrid. This substation is downstream of Transgrid's 330/132 kV Newcastle Bulk Supply Point (BSP), located approximately 6 km to the west. Figure 3-13 (see next page) provides a geographic diagram showing the location of Ausgrid and Transgrid's existing infrastructure.

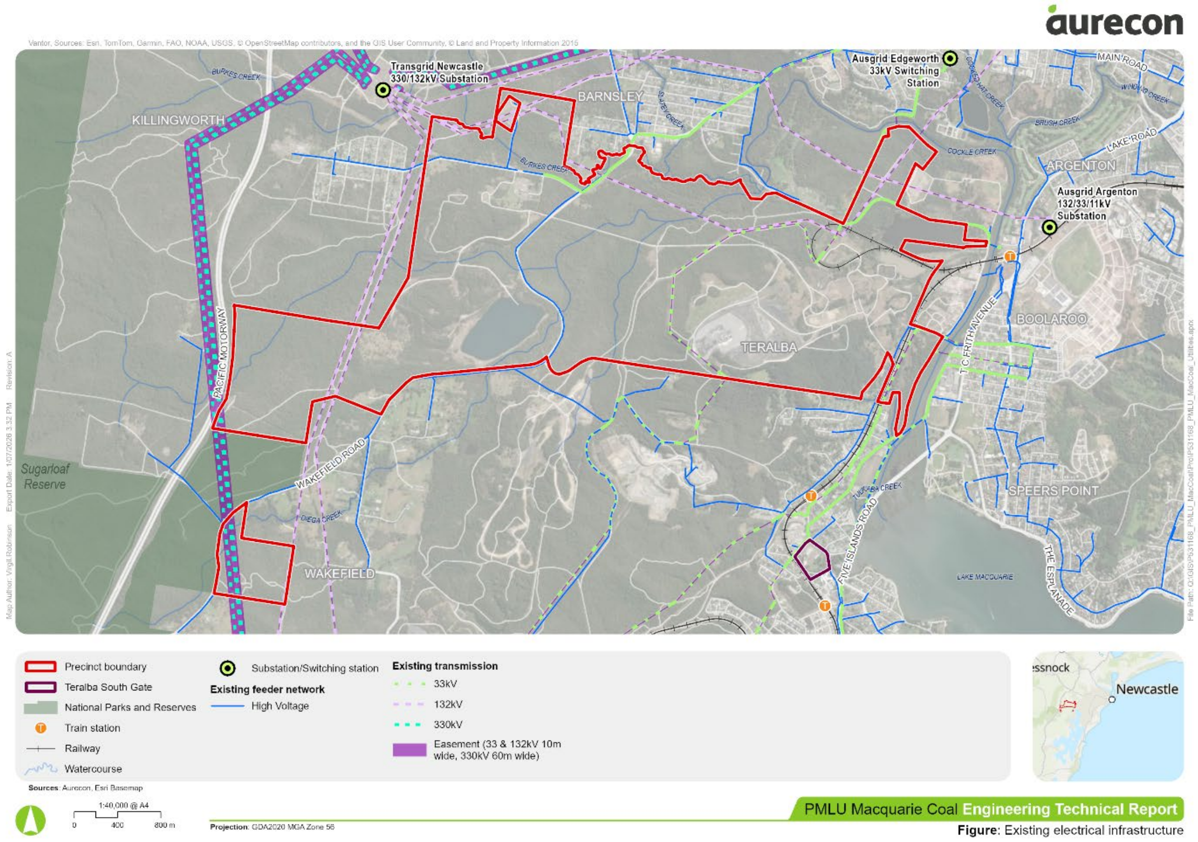

Figure 3-13 Existing Electrical Infrastructure. Note that 11 kV and low voltage infrastructure are also present but are not shown on this map.

3.4.1 Demand Assessment

The expected electricity demand of the Precinct and Teralba Southgate is outlined in Table 3-22. Electrical demand of the proposed Precinct and Teralba Southgate was calculated based on the Gross Floor Area (GFA) proposed by GHD. The GFA is multiplied by maximum demand by land use as outlined in Table 3-24 of the AS/NZS 3000:2018 standards (shown toward the end of Section 3.4.1) to produce the electrical demand estimates of this report.

Two scenarios are provided, a low case and a high case, along with corresponding post-diversity demand cases. Note that actual long-term demand will depend on how the Precinct and Teralba Southgate develops over time, and it is currently uncertain whether it will align more closely with the low-demand or high-demand scenario.

Table 3-22 Precinct and Teralba Southgate Expected Electricity Demand

Area	Name	Land Use Type	Gross Floor Area (GFA) (ha)	Low Case (MVA)	High Case (MVA)	Post Diversity Factor Electrical Demand (Low) (MVA)	Post Diversity Factor Electrical Demand (High) (MVA)
1	Cockle Creek	Heavy industrial	87.67	65.76	118.36	52.60	94.69
2	Killingworth	Light industrial	61.93	30.97	55.74	30.97	55.74
3	West Lake	Tourism / resort	16.09	8.04	17.70	6.43	14.16

Area	Name	Land Use Type	Gross Floor Area (GFA) (ha)	Low Case (MVA)	High Case (MVA)	Post Diversity Factor Electrical Demand (Low) (MVA)	Post Diversity Factor Electrical Demand (High) (MVA)
4	Teralba Southgate	Residential – flats/units	12.60	5.04	7.56	4.03	6.05
5	Barnsley West	Light industrial	6.38	3.19	5.74	3.19	5.74
Total*			**184.67**	**112.99**	**205.10**	**97.23**	**176.37**

**Totals may not sum due to rounding*

Adopted energy demand by land use assumptions are summarised in the table below.

Table 3-23 Assumed demand by land use, based on AS/NZS 3000:2018.

Land Use Type	Low Case (MVA/ha)	Low Case Assumption	High Case (MVA/ha)	High Case Assumption
Light industrial	0.5	Light and power and ventilation and air conditioning, using lower limit values	0.9	Light and power and ventilation and air conditioning, using upper limit values
Heavy industrial	0.75	50% greater than maximum demand for light industrial	1.35	50% greater than maximum demand light industrial
Tourism / resort	0.5	Assumed to be offices occupancy type, non air-conditioned, average light and power	1.1	Assumed to be offices occupancy type, average light and power, air-conditioned using zonal reheat (upper limit values)
Residential – flats/units	0.4	Aligned with typical DNSP benchmarks (sourced from: CitiPower & Powercor data)	0.6	Aligned with typical DNSP benchmarks (sourced from: CitiPower & Powercor data)

Several assumptions have been used to provide a conservative estimate of the energy demand, including:

- Diversity factors of 0.8 for residential and tourism / resort land use, and 1 for industrial land use.
- Yield demand estimates are based on gross floor area and does not differentiate between 1-, 2-, or 3-bedroom units delivered in the Precinct.
- Transmission capacities are based upon reported figures by Ausgrid and Transgrid.

Table 3-24 Maximum Demand – Energy Demand Method for Non-Domestic Installations (AS/NZS 3000:2018)

Type of occupancy		Energy demand	
		Range, VA/m^2	Average, VA/m^2
Offices	Light and power	40-60	50
	Airconditioning: ■ Cooling ■ Reverse cycle ■ Zonal reheat ■ Variable volume	 30-40 20-30 40-60 20	 35 25 50 20
Carparks	Open air EV charging Basement EV charging	0-10 5-15 10-20 10-30	5 10 15 20
Retail shops	Light and power Airconditioning	40-100 20-40	70 30
Warehouses	Light and power Ventilation Special equipment	5-15 5 (use load details)	10 5
Light industrial	Light and power Ventilation Airconditioning Special equipment	10-20 10-20 30-50 (use load details)	15 15 40
Taverns, licensed clubs	Total	60-100	80
Theatres	Total	80-120	100

Note: EV charging relates to charging equipment associated with electric vehicles and should be considered in addition to all other energy demands.

3.4.2 Residual Capacity of the Existing Infrastructure

The existing network consists of 132 kV transmission and 33 kV sub-transmission networks via several substations owned by Transgrid and Ausgrid. Most of these assets are located throughout the public road reserve, as shown in Figure 3-14.

Customer loads are supplied by 33 kV and 11 kV distribution feeders by various Sub-Transmission and Zone Substations surrounding the precinct, including the following:

- Argenton 132/33/11 kV Zone Substation
- Edgeworth 33/11 kV Switching Station

The thermal ratings and loadings of the 11 kV feeders are currently unknown and will require consultation with Ausgrid, supported by detailed planning studies.

Look Up and Live Map

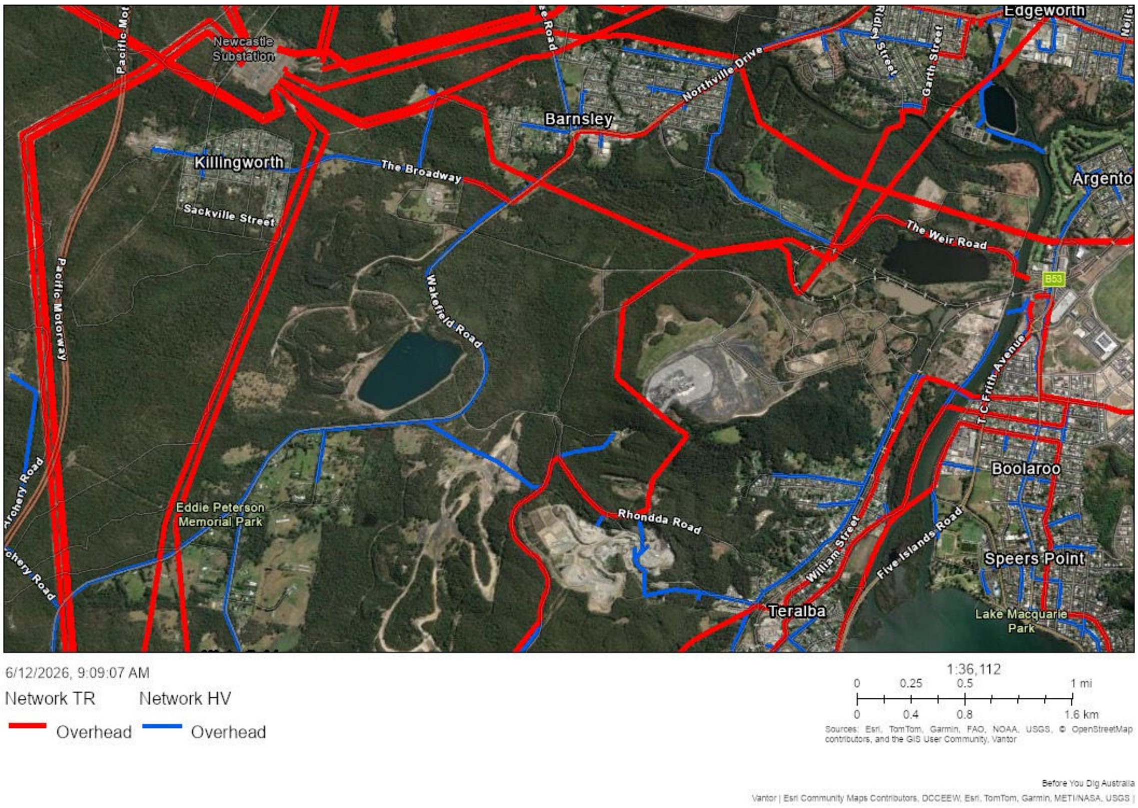

Figure 3-14 Surrounding Transmission Infrastructure

Based on Ausgrid's Rosetta Data Portal, the Argenton ZS has residual capacity of up to ~30 MVA at on the 11 kV bus. This capacity may accommodate part of the Precinct development until upgrade thresholds are reached. As this capacity is progressively utilised, the installation of additional 11 kV feeders, each with a typical capacity of approximately ~5 MVA, will likely be required to service incremental load. There may be opportunity to utilise existing 11kV feeders already throughout the site, however, the residual capacity of these assets is not known.

Beyond ~30 MVA, moderate to major upgrades may be required at Argenton ZS to increase the transfer capacity. This may include the addition of a new 132/11kV transformer and / or additional 132kV incoming feeder via Newcastle BSP.

It is noted that ~400 MW of load capacity is available at the 132 kV busbar at Transgrid's Newcastle Bulk Supply Point. The load transfer capability at this site is constrained by the three existing 330/132 kV transformers. Additional load may trigger the installation of a fourth 330/132 kV transformer, noting that provision for space and switchgear is already available.

3.4.3 Proposed Infrastructure and Staging

From the expected electricity demand of approximately 97 MVA and 176 MVA for a low and high case (Table 3-22), respectively, infrastructure upgrades will be required to unlock the Precinct's ultimate forecast electrical demand. Note that long-term energy needs may evolve over time, and the Precinct's demand requirements may track towards either the low or high demand scenario.

Large Energy Consumers

Large energy consumers such as data centres, smelters and heavy industries, should be located in the north-west section of the Precinct to leverage proximity to Newcastle BSP, whilst immediate / short term small loads can leverage the existing Argenton ZS. This approach aligns with high water and energy use requirements and supports efficient resource management across the Precinct.

Network Upgrades

For loads exceeding ~30 MVA, moderate to major substation upgrades are likely required, which would allow up to ~60 MVA of load. This is likely to include an additional 132/11kV transformer and relevant switchgear upgrades. Extension or additional 11kV feeders into the Precinct are also likely to be required. Cockle Creek (Area 1), which is intended to accommodate heavy industrial land uses (as outlined in Table 3-22), benefits from its proximity to the Argenton ZS. This may reduce the distance and extent of network augmentation required to service demand from this area.

For loads in the range of approximately 60-90 MVA or greater (the limit at Argenton ZS), the development of one or more new 132/11 kV zone substations, each with a capacity of ~60-90 MVA may be required to supply the ultimate Precinct load. These substations would likely be supplied from Transgrid's Newcastle BSP. The availability of ~400 MW of capacity at the 132 kV busbar has the potential to support this upgrade, noting that this may trigger the need for a fourth 330/132 kV transformer – depending on investment timing and potential competing loads in the region. Killingworth (Area 2) in the north-west sections and located closer to the Newcastle BSP may benefit from reduced transmission connection distances and corresponding reduction in the extent of network augmentation required to establish supply.

Another potential option includes building an additional 132 kV feeder between the Newcastle BSP and Argenton ZS. However, this approach may be constrained by technical and / or space limitations at both Argenton ZS and Newcastle BSP. Further, this upgrade is expected to be highly challenging as it would require substantial new easements and corridors.

Teralba Southgate in the southern section is anticipated to be supplied via existing network with 1-2 new 11 kV feeders from the nearest zone substation.

New connections into the network are assessed on a case-by-case basis in collaboration with Ausgrid and Transgrid and subject to capacity within the existing network.

Figure 3-15 shows the proposed infrastructure staging for electricity network upgrades. A summary of the staging of required upgrades to Ausgrid and Transgrid's network is outlined in Table 3-25 further below.

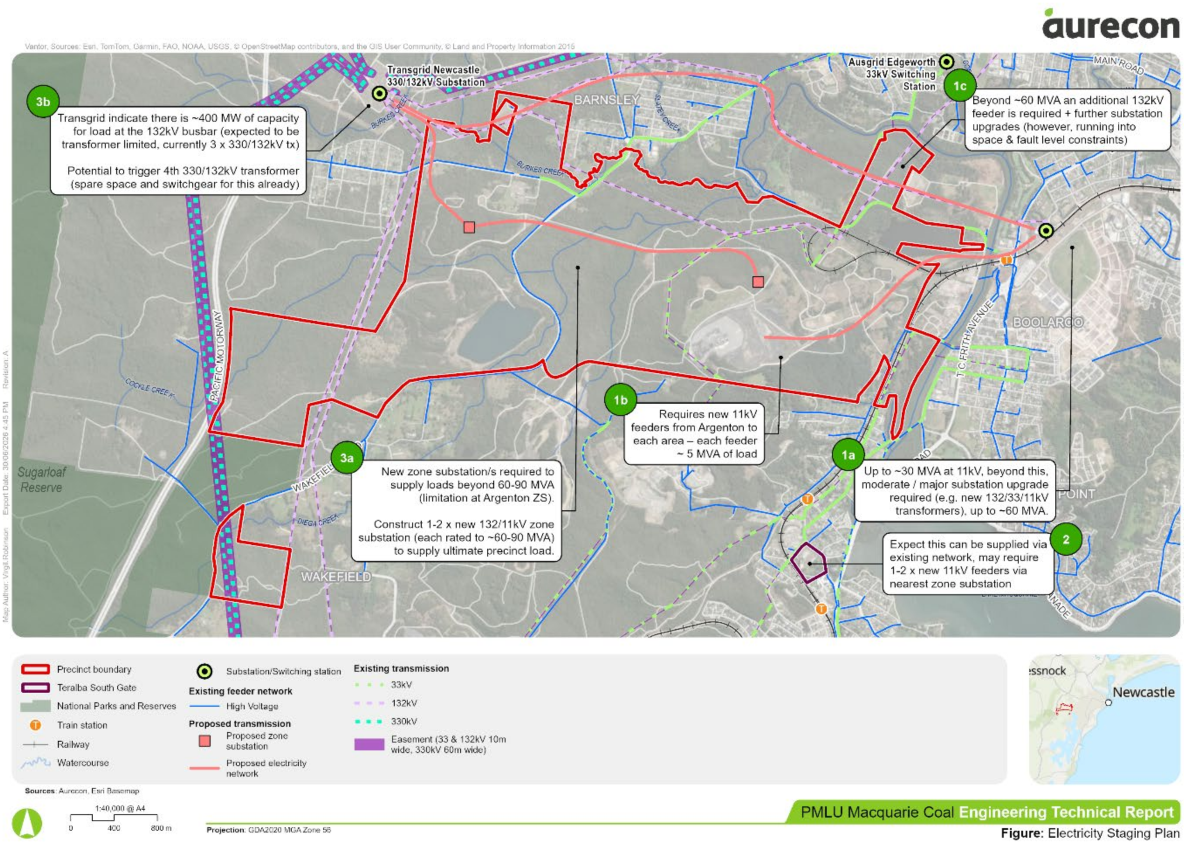

Figure 3-15 Proposed Infrastructure and Staging

Table 3-25 Precinct Electricity Network Upgrades Staging

Stage	Required Infrastructure Upgrades
Short- term: 0-5 years (up to ~30 MVA)	
1a	Utilise existing capacity at Argenton Zone Substation up to ~30 MVA at the existing busbar. Build 11kV network in accordance with capacity, with each feeder able to supply approximately 5 MVA. Prioritise utilising existing feeders before building out new infrastructure.
1b	Install new 11kV feeders from Argenton ZS to each area, with each feeder rated for ~5 MVA of load.
2	Area 4 (Teralba Southgate) is expected to be supplied via existing network, but may require 1-2x new 11kV feeders.
Medium term: 5-15 years (beyond ~30 MVA)	
1a	Beyond ~30 MVA at 11kV, moderate to major substation upgrades may be required, including new 132/33/11kV transformers. This may accommodate up to ~60 MVA.
1c (optional)	Beyond ~60 MVA, an additional 132kV feeder may be required, which may necessitate further substation upgrades. However, space and fault-level constraints may apply. Note that to install an additional 132kV feeder, areas for towers will need to be located. Tower location and clearance requirements present constraints, with above-ground expansion appearing to be limited. Underground installation with new circuit breakers is feasible but challenging due to substation size constraints. This option is optional for future consideration, as constructing a new zone substation may be a more cost-effective alternative.
Long-term: 15 years+ (beyond ~60 MVA)	
3a	Beyond ~60-90 MVA, new zone substation(s) may be required to supply loads (limitation at Argenton ZS). This may require construction of 1-2 new 132/11kV zone substations, each rated to approximately 60-90 MVA, to supply ultimate Precinct load.
3b	Based on Transgrid's Rosetta Data Portal, approximately ~400 MW of capacity is available at the 132kV busbar (expected to be transformer-limited, currently 3 x 330/132kV transformers). There is potential to trigger installation of a 4th 330/132kV transformer (with spare space and switchgear already available).

Long-term energy needs may evolve over time, and the Precinct's demand requirements may track towards either the low or high demand scenario. This uncertainty is not expected to constrain initial development, which is likely able to proceed with minimal network augmentation beyond additional feeders to service early-stage loads.

Data Centre Considerations

Data centres represent a significant class of high-demand electrical loads requiring dedicated infrastructure planning. These facilities should be evaluated within the broader high-voltage user strategy, with preference given to locations that maximise network efficiency and align with water and energy resource availability. Data centre placement should be assessed on a case-by-case basis in collaboration with Ausgrid and Transgrid, subject to capacity within the existing network and alignment with Precinct masterplan objectives.

Aurecon has based the required infrastructure upgrades for the delivery of Precinct on capacity data from Ausgrid and Transgrid's Rosetta Data Portal. However, due to changing connection load profiles in the network, as well as limited data availability on ratings, consultation with Ausgrid and Transgrid may be required to consolidate required infrastructure upgrades and relevant capital expenditure investments.

3.4.4 Constraints and Opportunities

Constraints

Development of transmission infrastructure within the Precinct must consider existing on-site infrastructure assets, including not encroaching on exclusion zones around transmission lines and associated structures. These include access and earthing, which may limit where infrastructure can be located.

For example, for transmission lines operating above 132kV, the following exclusion zones apply for Transgrid assets:

- Access exclusion zones are within 20m from any part of structure.
- Earthing exclusion zones are within 30m from any part of structure.
- Centre exclusion zones are within 17m either side from centre of the transmission line.

Figure 3-16 illustrates exclusion zone requirements within Transgrid's easements.

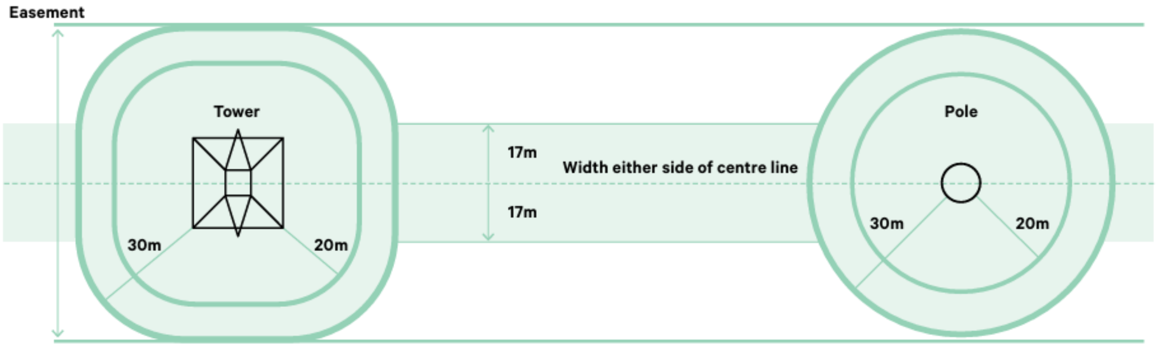

Figure 3-16 Transgrid's Easement Requirements for 220 kV and Above

Additionally, new network infrastructure will require dedicated easements to accommodate lines, substations and access. These easements may need to be incorporated into the Precinct layout and may reduce developable area or constrain land configuration. Early planning for these corridors is encouraged to avoid re-work.

Additionally, there is a long-term constraint associated with the potential need for a new zone substation within the Precinct. The siting of a ZS will need to be considered as part of the development footprint, including allowance for:

- Adequate land area and suitably topography
- Access for construction and ongoing maintenance
- Integration with existing and future network connections
- Appropriate buffers to surround land uses (e.g. noise, safety, visual considerations)

The inclusion of a ZS may have implications for land use zoning and the overall arrangement of development within the Precinct. As such, early planning of a suitable site is recommended.

All development within easements is subject to the requirements and approvals of the relevant network service provider.

Opportunities

While the servicing strategy for the Precinct has been set to meet satisfactory performance electricity reliability standards, there are still several opportunities to improve the electricity servicing of the expected development and particularly the sustainability of the Precinct development. These opportunities include:

- On-site renewable energy generation and/or coordinated microgrid could offset the need for some infrastructure upgrades. Given the extensive land area available, potential options to be assessed include rooftop / ground mounted solar and / or battery energy storage system (BESS). However, the scale of potential changes to the suggested utilities infrastructure scenarios will need to be determined following thorough assessment and sizing of potential renewable energy options.

- Transport electrification infrastructure – enabling the transition to electric, hybrid and alternative-fuel freight vehicles supports both environmental sustainability and electrical load management within the Precinct. Opportunities to integrate smart charging and energy management systems should also be explored to optimise demand profiles and align with on-site renewable energy generation. Future-proofing the development for charging and refuelling infrastructure will enable staged deployment as demand grows.
- Implementation of a community battery arrangement into the Precinct may provide a level of self-supply during outages, but more importantly help reduce electricity billing costs for end-users in the Precinct. This concept could provide resilience to the Precinct by providing islanding capability during outages. The batteries should be able to operate and supply at least a portion of energy to the Precinct during outages.
- It is noted that development activity is anticipated in the surrounding area, including densification around train stations. It is expected that growth in the area may expedite the timing of required infrastructure upgrades.

3.4.5 Planning framework and control recommendations

Further consultation with Ausgrid and Transgrid will be required to facilitate planning of the proposed upgrades in the Precinct. The final servicing strategy will be informed by Ausgrid and Transgrid's technical, planning and regulatory requirements.

It is recommended that the following controls are applied to developments within the Precinct to meet development requirements and enhance the efficiency of the electricity servicing network:

- Smart pole network infrastructure (incorporating street lighting, public Wi-Fi and CCTV technology)
- On-site rooftop solar panel systems
- Community battery network

3.5 Gas Servicing Strategy

3.5.1 Demand Assessment

A demand assessment for gas has not been undertaken as part of this proposal. Gas connections to the Jemena network would be made on a case by case basis.

3.5.2 Proposed Infrastructure

Future gas servicing for the Precinct is proposed to be established via connection to the existing Jemena gas network at the intersection of Racecourse Road and Blair Street, located to the south-east of the site.

This location represents the most viable connection point, comprising an existing 160 mm diameter gas main operating at approximately 210 kPa, which can be extended to service the Precinct subject to detailed feasibility assessment and coordination with Jemena.

The proposed servicing approach assumes a staged extension of the gas network from this connection point into the eastern portion of the Precinct, where proximity to the existing main minimises connection distance, enabling lower upfront infrastructure costs and improved commercial viability.

In line with this approach, gas-dependent land uses are anticipated to be prioritised within the eastern and Teralba Southgate areas of the Precinct, where connection feasibility is greatest. Extension of gas infrastructure to the western portions of the Precinct may be more constrained and would be subject to sufficient demand to justify the additional capital investment.

All gas connections will be subject to Jemena approval and are expected to be delivered on a case-by-case basis, informed by final land use, demand profiles and commercial considerations.

In the instance of manufacturing, specialist gasses are often used, these are traditionally procured through bottled services on a site-by-site basis.

Should a proponent wish to reticulate gas, it would be done by reticulating a trunk main at 160 mm diameter operating at approximately 210 kPa. Smaller mains would then branch out from this main.

3.5.3 Infrastructure Staging

No gas provision is proposed, should gas be required in the precinct it is expected to be extended from the corner of Racecourse Road and Blair Road.

3.5.4 SWOT Analysis

The strengths, weaknesses, opportunities and threats to the gas servicing of the proposed Precinct Master Plan are outlined in Table 3-26.

Table 3-26 Precinct Master Plan Gas Servicing SWOT Analysis

Strengths	Weaknesses
Clear and identifiable connection point to the regional gas network at Racecourse Road and Blair Street, providing a defined pathway for future supply if required. Teralba Southgate area has relatively good access to gas infrastructure, offering a logical early-use zone for any gas-reliant activities. Gas is not a dependency for the Precinct, allowing flexibility in planning and supporting a transition toward electrification and alternative energy solutions.	No existing gas infrastructure within the Precinct, requiring full extension of network to service development. Gas servicing is non-essential and subject to case-by-case commercial viability, creating uncertainty for long-term planning. Western portions of the Precinct are remote from viable connection points, increasing lead-in costs and reducing feasibility for distributed gas supply. Reliance on electrification for most users increases pressure on electrical infrastructure compared to a diversified energy mix.
Threats	**Opportunities**
High capital cost and uncertain take-up risk for gas network extension, particularly if demand is not concentrated or guaranteed. Stranded asset risk if gas infrastructure is delivered but broader policy, market or tenant preferences shift toward electrification and decarbonisation Large industrial users requiring gas may be constrained in siting, particularly in western areas where connection is less viable Increasing policy and market pressure toward net zero outcomes may limit the role of gas over the life of the Precinct, impacting long-term investment value	Strategic clustering of gas-dependent or industrial thermal users in the eastern Precinct to optimise proximity to the Blair Street connection and improve feasibility of a trunk main extension. Ability to avoid traditional gas dependency and instead position the Precinct as a low-carbon industrial hub through electrification, renewable energy integration and emerging energy technologies. Integration of precinct-scale alternative energy systems such as: ▪ Electrification of industrial processes supported by strong regional grid capacity ▪ Renewable energy (solar and storage) leveraging large industrial roof areas ▪ Hydrogen or clean fuel production aligned with regional initiatives Potential to plan energy infrastructure holistically, enabling distributed energy systems, microgrids or hybrid solutions rather than retrofitting gas later

	Opportunity to align land use planning with energy intensity, placing high energy users near electrical capacity and minimising reliance on gas altogether

3.5.5 Planning Framework and Control Recommendations

Jemena remain the primary gas network provider in the region. All new connection will be on a case by case basis and a commercial agreement. Future provision in the road corridor should be made for potential gas. No site specific controls are proposed for gas.

3.6 Telecommunication Servicing Strategy

Minor network augmentations are generally required to meet future demand and support development, including potential augmentation to the nearby fibre access nodes, underground pit and pipe and ducting works. However, these works are generally reactive and targeted as formal customer requests are submitted. They have minimal implications to be considered when developing the Precinct Master Plan as works are generally within the existing and proposed road network.

3.6.1 Demand Assessment

At this stage of planning, telecommunications and data demand across the Precinct is inherently difficult to quantify with accuracy, as ultimate requirements are highly dependent on the mix of future land uses and tenant profiles, particularly where high-demand users such as data centres, advanced manufacturing or logistics hubs may be introduced. While baseline mobile coverage and proximity to existing fibre networks provide a strong foundation, detailed demand profiles will evolve over time in response to specific development proposals and technology requirements, and are therefore typically addressed progressively as part of subdivision and development staging.

For greenfield precincts and subdivisions, connection to the National Broadband Network (NBN) generally follows a developer-led process, where the proponent engages directly with NBN Co to deliver pit and pipe infrastructure in accordance with NBN standards and design guidelines. This involves preparing a telecommunications servicing design, constructing the passive network (conduits, pits and pathways) as part of civil works, and coordinating installation of active fibre infrastructure by NBN Co once development thresholds are met. The process is typically aligned with subdivision approvals and staged development, with costs and delivery arrangements varying depending on development scale, staging and required service levels.

Notwithstanding the above, access to both the Telstra copper system and NBN (through fibre or satellite) are currently possible to enable development to occur.

3.6.2 Proposed Infrastructure

For telecommunications, the proposed infrastructure at a precinct scale is relatively light-touch and largely developer-driven, focused on enabling a scalable network rather than delivering major upfront trunk assets:

The telecommunications servicing strategy for the Precinct is expected to comprise staged delivery of fibre-ready passive infrastructure, including underground pit and pipe networks installed within road corridors as part of subdivision works. These conduits and pathways form the backbone for future fibre deployment and are typically designed and constructed by the developer in accordance with NBN Co requirements. Connection points are likely to be established from existing networks along Powerhouse Road to the east and The Broadway to the west, enabling initial unlocking of development areas, with further extensions delivered progressively as demand grows and new stages are brought online. The southern portions of the site would directly connect to an extension of the Broadway infrastructure. The Teralba Southgate site can make direct connection to the fibre immediately surrounding the site.

Active telecommunications infrastructure, including fibre optic cabling, exchanges, and network upgrades, would be delivered incrementally by service providers such as NBN Co, Telstra or other carriers in response to confirmed development demand. Large-scale or high-demand users, such as data centres or advanced industrial facilities, may require dedicated fibre connections or bespoke network solutions. At a precinct scale, additional infrastructure such as mobile towers, fibre backhaul upgrades and network redundancy may also be introduced over time to support reliability, capacity and future digital requirements, though these are typically reactive investments aligned with occupancy and user needs rather than upfront constraints on development.

The Precinct currently benefits from generally strong baseline 4G and 5G mobile coverage supported by existing telecommunications infrastructure in the surrounding area. However, the strategic scale and diversity of proposed land uses, including industrial, logistics and potential high-demand digital users, are likely to increase data traffic and network load over time.

As development intensifies, there may be a need for additional mobile towers or small cell installations within the Precinct to maintain service quality, improve coverage in more remote or topographically constrained areas, and meet increasing capacity demands. This is particularly relevant for large-format industrial estates where building footprints, equipment and operational activities can impact signal propagation. The implementation of additional mobile infrastructure would typically occur in response to demonstrated demand and be delivered in coordination with telecommunications carriers, with opportunities to integrate towers into the broader precinct design where possible.

Large-scale data users, such as data centres or high-intensity digital infrastructure, would be treated on a case-by-case basis due to their unique and often significant bandwidth, latency and resiliency requirements. These users are typically expected to deliver their own dedicated fibre connections and backhaul infrastructure directly to major network nodes or exchanges, rather than relying solely on the shared precinct network. This approach ensures that their operational requirements can be met without placing undue demand on the broader precinct telecommunications infrastructure, while also allowing bespoke solutions aligned with global standards for redundancy, security and performance.

3.6.3 Infrastructure Staging

Initial NBN staging for the Precinct would follow a phased, developer-led rollout aligned with early development areas and existing network proximity:

- **Stage 1 enabling connections**
  Initial connections would be established from existing telecommunications corridors at Powerhouse Road (east) and The Broadway (west), which represent the nearest viable tie-in points to existing fibre and copper infrastructure. These connections would support early activation of Areas 1, 2 and initial industrial or commercial uses, enabling development to proceed without major upfront upgrades.
- **Early passive infrastructure delivery**
  Developers would construct pit and pipe (conduits and pathways) within road corridors as part of subdivision civil works, forming the backbone for fibre rollout. This infrastructure is sized to accommodate future capacity, avoiding rework as demand increases.
- **Incremental fibre rollout by NBN Co**
  Once development thresholds are met, NBN Co would progressively install active fibre infrastructure to service connected lots and buildings. This ensures that capital investment aligns with actual demand rather than requiring full precinct servicing upfront.
- **Staged expansion deeper into the precinct**
  As development expands south and west, the network would be extended from the initial tie-in points, with additional conduits installed to service later stages. More remote areas, particularly those further from The Broadway corridor, would be progressively connected as demand justifies the extension.

3.6.4 SWOT Analysis

The strengths, weaknesses, opportunities and threats to the telecommunications servicing of the proposed Precinct Master Plan are outlined in Table 3-27.

Table 3-27 Precinct Master Plan Telecommunications Servicing SWOT Analysis

Strengths	Weaknesses
Good baseline 4G and 5G mobile coverage across the site Proximity to existing fibre networks and multiple connection points (east and west) Established ability to enable early development with minimal upfront infrastructure Combined with water and electricity, the north western portions of the site appear suited to data intensive industries.	Limited telecommunications infrastructure currently within the site Data demand is uncertain and difficult to quantify at this stage Significant reliance on staged, developer-led delivery and external providers (NBN, Telstra, carriers) Southern portions of the site along Wakefield Road have very poor fibre connectivity.
Threats	**Opportunities**
Risk of under provisioning early infrastructure if demand grows faster than anticipated Large-scale users (e.g. data centres) placing pressure on network capacity if not managed separately Dependence on third-party providers for timing and delivery of active infrastructure Potential coverage or capacity gaps emerging in more remote or heavily developed industrial areas without targeted upgrades	Scalable, fibre-ready pit and pipe network can be rolled out efficiently with development staging Ability to design a high-quality digital precinct from the ground up, including redundancy and smart infrastructure Potential to integrate additional mobile towers, small cells and fibre backhaul as demand grows Opportunity to accommodate high-value digital users, with tailored solutions for data-intensive industries

3.6.5 Planning Framework and Control Recommendations

The following planning and control measures are recommended (but not ultimately required) to support a scalable, resilient and high-quality telecommunications network across the Precinct:

- **Fibre-ready subdivision design**
  Require all new developments to deliver pit and pipe (conduit and pathway) infrastructure within road corridors in accordance with NBN Co guidelines, ensuring the precinct is “fibre-ready” from the outset and avoids costly retrofits.
- **Early engagement with NBN Co and carriers**
  Mandate early coordination with NBN Co and telecommunications providers during planning and subdivision stages to confirm connection pathways, servicing strategies and staging requirements.
- **Provision for multiple connection points and redundancy**
  Encourage network designs that incorporate multiple fibre entry points (e.g. east and west connections) and pathways to support redundancy, improve resilience, and reduce outage risk for critical users.
- **Corridor protection and utility coordination**
  Safeguard telecommunications corridors within road reserves and coordinate with other utilities to ensure adequate space for future fibre, ducts, and upgrades without conflicts.
- **Allowances for future mobile infrastructure**
  Embed planning controls that enable additional mobile towers or small cells to be integrated within the

precinct (e.g. within industrial lots, road reserves or shared infrastructure), subject to demand and carrier requirements.

- **Support for high-demand users**
  Identify and zone suitable areas for high-data users (e.g. data centres, logistics hubs) and require these developments to provide dedicated fibre connections and backhaul solutions where needed.
- **Smart infrastructure integration**
  Encourage incorporation of digital infrastructure such as smart poles, IoT systems, and public connectivity (e.g. Wi-Fi, sensors) to support future-ready precinct operations.

3.7 Waste Servicing Strategy

Waste servicing for the Precinct and Teralba Southgate is dependent on the volume and types of waste expected to be generated by the development. A demand assessment was performed to consider whether there are any major issues or opportunities associated with the proposed land uses that may necessitate enabling infrastructure to be built into the design or restrict the types of development that can be considered.

It is noted that the demand values provided below are based on broad assumptions for land use (see Section 3.1), intended to provide order of magnitude estimates to help inform planning decisions. These assumptions, calculations and findings should be revised as further detail is obtained.

3.7.1 Demand Assessment

There are three key stages relevant to the development of the Precinct and Teralba Southgate that are expected to generate significantly different types and volumes of waste, being:

- Site clearance and preparation
- Construction
- Ongoing operations.

Each stage requires consideration to evaluate capacity in the local region and support development of appropriate waste servicing strategies.

Site clearance and preparation

This stage of development is expected to generate green waste through land clearing. However, this is manageable on site through mulching and chipping, generating a product that either holds commercial value, or can be disposed of appropriately on site or managed through commercial contracts with organic waste processors. This is not considered an issue for the development.

Should there be contamination issues on site (see the accompanying Contamination Engineering Report), this may result in specific management requirements. It is understood that waste management facilities in the region have licenses to manage certain issues (e.g., lead contamination) but this would need to be investigated and appropriately managed. It has been assumed for the purposes of this analysis that any land contamination issues would be resolved locally and would not have any significant impact on waste infrastructure; this assumption should be revisited as further information on this issue is obtained.

Construction

Building construction is a major short-term generator of waste, producing various waste streams including general waste that requires landfilling, and a range of materials that can be recycled including packaging (paper and cardboard), metals, concrete, bricks and rubble. Other streams that may be generated include hazardous waste and dangerous goods.

The volume of waste generated during construction is highly variable, depending on what is developed. It tends to be generated in large volumes over relatively short timeframes, and is typically managed through contracts with industrial waste management facilities. High-level estimations of construction waste expected to be generated by the development of the Precinct and Teralba Southgate in line with proposed land use scenarios by Area are presented in Table 3-28 (rounded to the nearest hundred tonnes). Estimates are based on an assumed building type aligned to rates provided in the Sutherland Shire Council's Waste Management Plan for construction waste[1], which is considered a suitable reference guide as there is no equivalent for the Lake Macquarie or Newcastle Council areas.

[1] Sutherland Shire Council (n.d.). Waste Management Plan: Demolition, Construction and Use of Premises

Table 3-28 Estimated Demand for Key Waste Streams Generated from Construction

Area	Assumed building type	Landfill (tonnes)	Mixed recyclables (tonnes)
1	Factory	1,400	7,700
2	Factory	1,000	5,400
3	Factory	300	1,400
4	Block of flats	800	4,600
5	Factory	100	600
	Total	**3,600**	**19,700**

Ongoing operations

Following construction, operational waste requires ongoing management. This includes:

- **Industrial waste**, for example from industrial and advanced manufacturing, data centres and digital infrastructure, freight logistics warehousing and distribution, and energy transition and renewable infrastructure
- **Commercial waste,** including from offices, retail outlets, food production and tourism
- **Residential domestic waste** from low-, mid- or high-density housing.

The volumes and types of waste vary significantly both between and within these groups. For example, within industrial waste, manufacturing is expected to generate large volumes of waste that may include hazardous and dangerous goods, depending on what is being manufactured, while operational waste from data centres and renewable energy infrastructure is likely to be minimal. For commercial waste, retail outlets could produce high volumes of cardboard, while food outlets might generate waste oil and organics.

The key waste stream from a local management perspective is products requiring landfilling, as landfills can be space constrained, have specific licensing requirements and require long-term planning to maintain capacity to support future development and population growth. Recyclable and organic waste represent a lesser issue particularly in high-density areas because there are typically private operators that can separate and recycle, reuse or otherwise process these materials. Hazardous waste and dangerous goods can be challenging to manage but are typically generated in small quantities and can be accepted by specialist facilities.

Similarly to construction waste, a high-level estimate of waste that could be expected to be generated by the Precinct based on the proposed land uses is provided in Table 3-29 (rounded to the nearest hundred tonnes per annum [p.a.]), to support identification of any key constraints that may limit the types of development that could be considered. Commercial and industrial areas (Areas 1, 2, 3 and 5) are based on waste generation rates from the NSW EPA[2] while residential waste (Area 4) applies rates based on recent survey data for Lake Macquarie.[3]

Table 3-29 Estimated Demand for Key Waste Streams from Ongoing Activities

Area	Assumed land use	Landfill (tonnes p.a.)	Mixed recyclables (tonnes p.a.)	Organics (tonnes p.a.)
1	Manufacturing	28,800	144,000	-
2	Wholesale trade	4,500	18,100	-
3	Accommodation & food services	13,500	7,600	7,600
4	Residential	500	200	500
5	Wholesale trade	500	900	-
	Total	**47,800**	**170,800**	**8,100**

[2] NSW EPA (2015). Pilot generator site-based audit: Commercial and industrial waste stream in the metropolitan levy areas of NSW
[3] NSW EPA (2026). Local Government Waste Data Survey 2024-25

3.7.2 Proposed Infrastructure

There are no known waste management facilities operating within the Precinct or Teralba Southgate boundaries. However, based on a review of facilities within 50km of the site and engagement with the Lake Macquarie City Council, there appears to be capacity to manage the anticipated volumes and types of waste in the broader region.

Existing infrastructure

Both the Precinct and Teralba Southgate fall within the borders of the Lake Macquarie City Council. Engagement with the Lake Macquarie City Council indicated that the Awaba Waste Management Facility would have capacity to accept new domestic waste streams from these sites. It also has the required licenses to accept specific hazardous waste types including asbestos and lead-contaminated soil. This facility can accept additional waste for a fee, but this is set at a less competitive price point compared to nearby alternatives, to maintain its focus on domestic waste.

It is understood that the nearby Summerhill Waste Management Centre is a large site with long-term capacity under its master planning. It is considered the biggest commercial landfill in the region, regularly entering into agreements with large waste producers.

In addition to the Council facilities, there are numerous private operators that can manage large volumes of commercial and industrial waste. Facilities accepting construction and operational waste include Newcastle Recycling Centre, Boral Recycling Kooragang, Concrush, Veolia Raymond Terrace Resource Recovery Park and Recycle Central Lake Macquarie. Cleanaway has a depot at Kooragang for industrial waste including hazardous materials and dangerous goods, although a new industrial development generating significant volumes of these types of waste would require scrutiny, due to the lack of regional infrastructure to support its management.

Based on the availability of existing infrastructure in the region as outlined above, no new infrastructure is proposed.

Legislation and state policy

It is noted that any future development would be subject to relevant waste legislation including the *Protection of the Environment Operations Act 1997*, *Waste Avoidance and Resource Recovery Act 2001*, *Environmental Planning and Assessment Act 1979, Environmentally Hazardous Chemicals Act 1985* and *Contaminated Land Management Act 1997*, and associated regulations.

In addition, developments should align with the policies and strategies adopted by the NSW Government. The NSW EPA's 2019 *Circular Economy Policy Statement* recognises the long-term economic, social and environmental benefits created the circular approaches that maximise the value of the resources and contribute to innovation and job creation. The *NSW Waste and Sustainable Materials Strategy 2041* sets targets that align with the National Action Plan, including to have an 80% average recovery rate from all waste streams by 2030 and significantly increase the use of recycled content by governments and industry. It is supported by $356 million in funding to ensure the NSW Government can deliver on its objectives.

3.7.3 Constraints and Opportunities

Constraints

Solid waste disposal requires trucks to collect and transport the waste to the processing facility. With no infrastructure already existing within the Precinct, this would increase traffic on surrounding suburbs, although this is not anticipated to be significant relative to current traffic flows.

Opportunities

Council and private waste management and recycling facilities are available in the local region to support waste processing for the Precinct and Teralba Southgate. The increase in supply of recyclable materials, in particular, may help strengthen demand for recycling and repair / reuse services and encourage growth of these industries.

There is also opportunity to design the Precinct in a manner that supports waste minimisation by promoting industrial symbiosis, such as circular flows of energy and materials, and coordination of waste management within industrial hubs to manage the impact on local traffic, and support waste separation to maximise resource recovery and minimise landfilling.

3.7.4 Planning Recommendations

Further engagement with local Councils and private waste management contractors will be required as planning for the Precinct and Teralba Southgate progresses and further detail on proposed developments, expected waste profiles and timings have been identified, to confirm residual capacity of their facilities.

As part of future planning stages, it is recommended that the Precinct design incorporates:

- Common service corridors, and other shared services, to minimise the impact on local residents
- Consideration of synergies between industrial developments, to support embedding of circular economy principles including flows of energy and materials (e.g., waste heat, resource recovery).

4 Conclusion

This Utilities Engineering Report demonstrates that the Macquarie Coal Complex Precinct can be effectively serviced to support its transition to a post-mining land use, providing a strong foundation for the proposed rezoning. The assessment confirms that, despite legacy constraints associated with historic mining operations and fragmented infrastructure, the Precinct benefits from key enabling factors including strong regional electricity connectivity, proximity to existing utility networks, and access to substantial on-site water resources. These attributes allow for a feasible “business as usual” servicing pathway to be established, demonstrating that the Precinct can be developed in a manner consistent with standard authority requirements and development servicing frameworks.

Two of the most significant constraints to servicing the Precinct relate to the absence of existing wastewater infrastructure and the physical barrier presented by Cockle Creek.

A key constraint is that the majority of the Precinct, is not currently connected to the Hunter Water wastewater network and historically relied on septic systems during its mining operations. This means that any future development will require the delivery of entirely new wastewater infrastructure, including pump stations and rising mains, to connect to regional treatment facilities such as Edgeworth Wastewater Treatment Works. While the site benefits from proximity to this plant, it is already forecast to reach capacity by 2029, with upgrades planned but not yet confirmed or funded. This creates both a timing and delivery risk, with development potentially constrained by the need for significant upfront infrastructure investment and reliance on external authority upgrades.

In addition, Cockle Creek represents a major physical and servicing barrier across the Precinct. The creek effectively separates the site from existing utility networks and requires new infrastructure crossings to enable both potable water and wastewater connections. While such crossings are technically feasible, they introduce increased cost, complexity and approval requirements, particularly where trenchless construction methods or environmental considerations are required. As a result, Cockle Creek not only influences the cost and feasibility of traditional servicing solutions, but also reinforces the need to consider alternative approaches such as decentralised systems, staged servicing, or precinct-scale infrastructure to reduce reliance on crossing this constraint in the early stages of development.

Importantly, the report also identifies that the Precinct is uniquely positioned to move beyond traditional servicing approaches and explore more innovative, precinct-scale solutions. Opportunities such as alternative water supply, decentralised systems, renewable energy integration, electrification, and scalable telecommunications infrastructure provide a pathway to enhance resilience, reduce infrastructure burdens, and align with broader sustainability and decarbonisation objectives. In this context, the proposed servicing strategy not only supports the immediate requirements of rezoning, but also establishes a flexible framework that enables future refinement and the progressive adoption of alternative servicing solutions as development demand, technology, and market conditions evolve.

Document prepared by

Aurecon Australasia Pty Ltd
ABN 54 005 139 873
Level 11, 73 Miller Street
North Sydney 2060 Australia
PO Box 1319
North Sydney NSW 2059
Australia

T +61 2 9465 5599
F +61 2 9465 5598
E sydney@aurecongroup.com
W aurecongroup.com

aurecon
Bringing ideas to life